Predicting Airline Passenger Satisfaction: A Comparative Evaluation of Logistic Regression and Gradient Boosted Trees Using KNIME

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Module: IJC428 Data Mining and AI
Word Count: 3,000

1. Abstract

Background: Airline passenger satisfaction has become a critical source of competitive advantage as the industry recovers from the unprecedented financial losses caused by the COVID-19 pandemic. The ability to accurately predict passenger satisfaction enables airlines to allocate service resources more effectively and implement targeted retention strategies.

Aim: This study aimed to predict airline passenger satisfaction using binary classification and identify the key drivers of satisfaction from a dataset of 5,196 passenger records containing 21 predictor variables.

Methods: Two supervised learning algorithms, logistic regression (LR) and gradient boosted trees (GBT), were implemented and compared in KNIME 5.3.3. Both models underwent 5-fold cross-validation and systematic hyperparameter tuning. Model performance was evaluated using accuracy, F1-score, AUC-ROC, and Cohen's Kappa.

Results: GBT outperformed LR across all metrics on the test set, achieving accuracy of 0.90, F1-score of 0.89, AUC-ROC of 0.97, and Cohen's Kappa of 0.80, compared to LR's corresponding values of 0.84, 0.82, 0.91, and 0.68. LR coefficient analysis identified travel type, cardholder status, and digital service quality as the strongest predictors of satisfaction.

Conclusion: GBT is the superior model for this classification task, requiring no imputation, standardisation, or variable removal. Findings provide actionable insights for airline management, though replication on larger datasets and incorporation of additional features such as in-flight Wi-Fi are recommended for future research.

2. Introduction

The airline industry has suffered unprecedented economic losses as a result of the COVID-19 pandemic. According to Sun et al. (2022), airlines worldwide suffered revenue losses of $373 billion USD in the year 2020 alone. In fact, Gudmundsson et al. (2021) found that it may take until this year for pre-pandemic passenger levels to return. Furthermore, the airline industry is one of the most competitive service industries in the world, with high levels

of passenger satisfaction identified as a primary source of competitive advantage (Soklaridis et al., 2024). Additionally, higher levels of passenger satisfaction have been shown to positively influence customer loyalty and the overall financial performance of airlines (Eshaghi et al., 2024). Therefore, the ability to understand and predict passenger satisfaction has never been more strategically paramount for airlines seeking to rebuild their customer base and restore financial stability.

This study frames this problem as a binary classification task, predicting whether a passenger is 'satisfied' or 'neutral/dissatisfied'. A tabular dataset of 5,196 passenger records and 21 independent variables was utilised, including passenger demographics, in-flight service ratings (e.g., food and drink, entertainment, seat comfort), and operational data such as flight distance and arrival delays. This study implemented two supervised learning classification models: gradient boosted trees and logistic regression. Their predictive performance was evaluated and compared by utilising KNIME software. The models developed in this study enable airlines to predict passenger satisfaction from demographic, service, and operational data, providing a practical tool for proactive satisfaction management. The findings further provide airline management with actionable insights into the key drivers of passenger satisfaction, enabling more targeted service improvements, loyalty programme strategies and operational decisions that support financial recovery in the post-pandemic aviation landscape.

3. Data Mining Theory

This study framed airline passenger satisfaction prediction as a binary classification task. Binary classification is appropriate when the dependent variable consists of two categorical outcomes and the objective is to predict class membership (Geron, 2023; James et al., 2014). Two supervised learning algorithms were selected: gradient boosted trees (GBT) and logistic regression (LR).

3.1 Logistic Regression:

Logistic regression was selected as the first method and interpretability benchmark for this study. It models the probability of class membership by passing a linear combination of predictors through the sigmoid function (Zhang et al., 2022):

$$P(Y=1|x) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)})$$

Each coefficient measures the change in log-odds of passenger satisfaction associated with a unit change in the predictor (James et al., 2014), providing direct insight into the direction and relative importance of each variable (Arshad et al., 2023). This interpretability is especially valuable as KNIME 5.3.3 does not display GBT feature importance scores. LR has demonstrated consistent performance in prior airline satisfaction studies, with Mittal et al. (2026) reporting an accuracy of 0.88 and an F1 of 0.86, and Zhang (2024) reporting an accuracy of 88% and an AUC-ROC of 0.94. However, LR was outperformed by tree ensemble methods in both studies, providing the rationale for selecting GBT as the second method.

3.2 Gradient Boosted Trees:

GBT is an ensemble learning algorithm that builds a strong predictive model by sequentially training decision trees, with each successive tree trained to reduce the prediction error of the previous one (Rizkallah, 2025):

$$F_m(x) = F_{m-1}(x) + \nu h_m(x)$$

where $F_m(x)$ is the updated model at iteration m, $h_m(x)$ is the new decision tree fitted to the residuals of $F_{m-1}(x)$, and η is the learning rate. The learning rate scales each tree's contribution and reduces overfitting (Natekin & Knoll, 2013).

GBT was selected as it directly addresses four structural limitations of LR, given this dataset's characteristics. First, LR assumes a linear relationship between predictors and log-odds (Ren et al., 2014), whereas GBT uses threshold comparisons and can capture complex non-linear relationships between predictors and satisfaction (Hastie et al., 2009). Second, LR is sensitive to multicollinearity (Luqman et al., 2025). The strong correlation between departure and arrival delay ($\rho \approx 0.75$) necessitated the removal of departure delay from the LR pipeline, whereas GBT is robust to correlated predictors through built-in feature selection, retaining both variables (Li et al., 2022). Third, LR requires complete data to calculate coefficients (Das et al., 2010), meaning imputation was required across several variables with

substantial missingness, which can introduce bias (Meng, 2025). GBT handles missing values natively by directing affected observations to the optimal child node during splitting (Chen & Guestrin, 2016). Fourth, LR requires z-score standardisation as coefficient magnitudes depend critically on input scales (Arafat et al., 2025), whereas GBT is unaffected by scale differences between variables such as flight distance (56–4,983 miles) and age (7–80 years) (Pinheiro et al., 2025).

The primary disadvantage of GBT is susceptibility to overfitting if tree depth or number of trees is too large (Zhang & Yin, 2023), which was addressed through 5-fold cross-validation and hyperparameter tuning. Prior studies confirm GBT's suitability: Zhang (2024) and Ramadhan & Putrada (2023) both reported XGBoost as the strongest-performing model, achieving accuracies of 95% and 96%, respectively. Zhang (2024) additionally reported a superior AUC-ROC of 0.99 for XGBoost, while Ramadhan & Putrada (2023) recorded the highest F1-score of 0.96.

Table 1

3.3 Alternative Methods Considered:
GBT were selected over random forest as gradient boosted variants have been shown to outperform random forest on comparable airline satisfaction datasets (Ramadhan & Putrada, 2023; Zhang, 2024). Neural networks were also considered, but GBT methods have been found to outperform neural networks and require less computational power on tabular data (Yildiz & Kalayci, 2025).

3.4 Performance Metrics:

Four complementary metrics were selected to evaluate model performance, as each measures a different aspect of classification quality. Accuracy measures the proportion of correctly classified instances:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

However, accuracy can be misleading with imbalanced class distributions (Tharwat, 2021). Given the mild class imbalance (56.3%/43.7%), three additional metrics were used. The F1-score uses the harmonic mean of precision and recall:

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$$

Ensuring strong performance across both satisfaction classes (Tharwat, 2021). AUC-ROC considers all possible decision thresholds, making it robust to class imbalance (Li, 2024). Finally, Cohen's Kappa accounts for correct predictions occurring by chance:

$$\kappa = (P_o - P_e) / (1 - P_e)$$

where P_o is observed accuracy and P_e is the expected chance agreement (McHugh, 2012).

Table 2

3.5 Improvements to Prior Studies:
This study improves on the previous studies mentioned above in three respects. Firstly, while Zhang (2024) relied on single train/test splits, this study applied 5-fold cross-validation to both models, which produces more stable and generalisable performance metrics and reduces the risk of overfitting (Abedin et al., 2026). Secondly, none of the prior studies utilised F1-score, accuracy, and AUC-ROC in combination (Mittal et al., 2026; Ramadhan & Putrada, 2023; Zhang, 2024). Thirdly, none of these studies utilised Cohen's Kappa. This study reports all four metrics, providing a more rigorous evaluation of classification performance.

4. Data Exploration and Preparation

4.1 Dataset Overview and Descriptive Statistics:

The airline passenger satisfaction dataset consists of 5,196 observations and 21 predictor variables. The dependent binary categorical variable is passenger satisfaction. There are twelve ordinal categorical service rating variables (on a scale of 0-5 or 1-5), five nominal categorical variables, and four continuous variables. Table 3 showcases all of the variables in

the dataset, along with their corresponding variable type, scale, and missingness percentage. Table 4 provides descriptive statistics for the four continuous variables, and Table 5 provides them for the twelve ordinal categorical service rating variables.

Table 3: *An Overview of the Dataset and Its Variables*

Note: Missingness is bolded when it is > 10% of the dataset.

Feature	Feature Type	Scale	Missing (n)	Missing (%)
Gender	Nominal Categorical	• Male • Female	0	0.00%
Customer Loyalty	Nominal Categorical	• Loyalty Cardholder • Non-Cardholder	0	0.00%
Age	Continuous	Years	0	0.00%
Type of Travel	Nominal Categorical	• Business Travel • Personal Travel	0	0.00%
Class	Nominal Categorical	• Business • Economy • Economy Plus	0	0.00%
Online Check-in	Nominal Categorical	• 0: No • 1: Yes	**1,531**	**29.46%**
Flight Distance	Continuous	Miles	**747**	**14.38%**
Departure/Arrival Time Convenience	Ordinal Categorical	0–5 (0 being not applicable: 1-5 with 1 being poor and 5 being excellent)	0	0.00%
Ease of Online Booking	Ordinal Categorical	0–5 Service Rating Scale (0 being not applicable: 1-5 with	0	0.00%

Feature	Feature Type	Scale	Missing (n)	Missing (%)
		1 being poor and 5 being excellent)		
Gate Location	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Food and Drink	Ordinal Categorical	0–5 Service Rating Scale (0 being not applicable: 1-5 with 1 being poor and 5 being excellent)	0	0.00%
Seat Comfort	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Inflight Entertainment	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
On-Board Service	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Leg Room Service	Ordinal Categorical	0–5 Service Rating Scale (0 being not applicable: 1-5 with 1 being poor and 5 being excellent)	0	0.00%
Baggage Handling	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Check-in Service	Ordinal Categorical	1–5 Service Rating Scale	0	0.00%

Feature	Feature Type	Scale	Missing (n)	Missing (%)
		(With 1 being poor and 5 being excellent)		
Inflight Service	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Cleanliness	Ordinal Categorical	1–5 Service Rating Scale (With 1 being poor and 5 being excellent)	0	0.00%
Departure Delay (min)	Continuous	Minutes	**747**	**14.38%**
Arrival Delay (min)	Continuous	Minutes	15	0.29%
Satisfaction	Nominal Categorical **(Dependent Variable)**	• Neutral or Dissatisfied • Satisfied	0	0.00%

Table 4: *Descriptive Statistics for the Continuous Variables*

Note: Bolded skewness indicates that it is > 2 (substantial skewness)

Feature	Min	Max	Mean	Median	SD	IQR	Skewness
Age	7	999	146.85	43	302.43	29–56	**2.46**
Flight Distance	56	4,983	1,163.74	825	977.82	406–1,662	1.16
Departure Delay (min)	0	565	14.35	0	35.10	0–12	**5.22**
Arrival Delay (min)	0	586	14.80	0	35.47	0–13	**5.24**

Table 5: *Descriptive Statistics for the Ordinal Service Rating Variables*

Feature	Scale	Min	Max	Median	IQR	Mode
Departure/Arrival Time Convenience	0–5	0	5	3	2–4	4
Ease of Online Booking	0–5	0	5	3	2–4	2
Gate Location	1–5	1	5	3	2–4	3
Food and Drink	0–5	0	5	3	2–4	4
Seat Comfort	1–5	1	5	4	2–5	4
Inflight Entertainment	1–5	1	5	4	2–4	4
On-Board Service	1–5	1	5	4	2–4	4
Leg Room Service	0–5	0	5	4	2–4	4
Baggage Handling	1–5	1	5	4	3–5	4
Check-In Service	1–5	1	5	3	2–4	4
Inflight Service	1–5	1	5	4	3–5	4
Cleanliness	1–5	1	5	3	2–4	4

4.2 Exploratory Data Analysis:

4.2.1 Boxplots of Ordinal Service Ratings:

Boxplots of the ordinal service ratings and Table 5 showed that the majority of the in-flight services had higher median ratings (median = 4) than the ground and booking services (median = 3). Figures 1–4 identified zero values in four service rating variables, confirmed as 'not applicable' responses rather than genuine poor ratings. These were recoded to missing values prior to partitioning.

Figure 1: Boxplot of the *Distribution of Departure/Arrival Time Convenience Ratings (median = 3, IQR = 2-4, Scale = 0-5)*

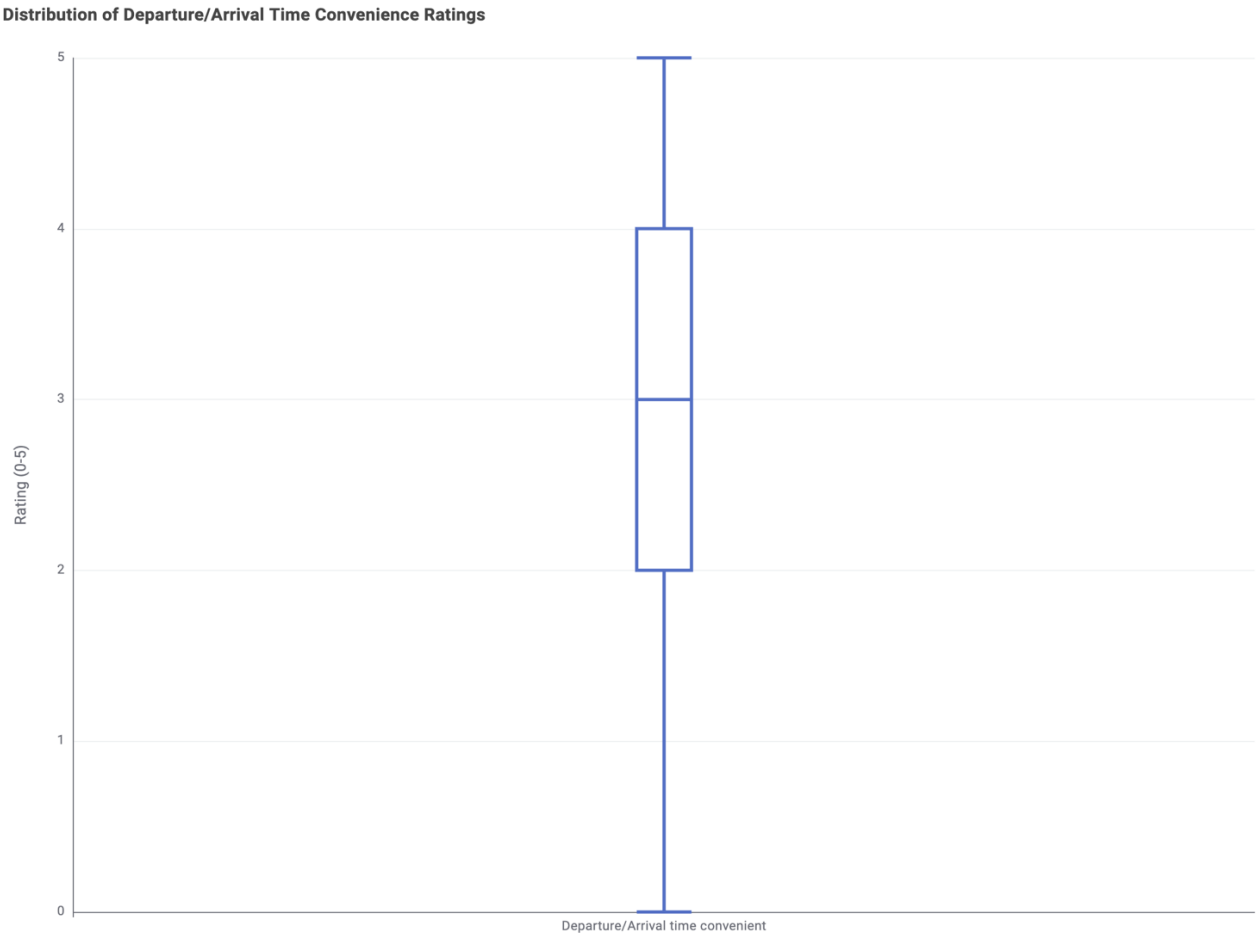


Figure 2: Boxplot of the *Distribution of Ease of Online Booking Ratings (median = 3, IQR = 2-4, Scale = 0-5)*

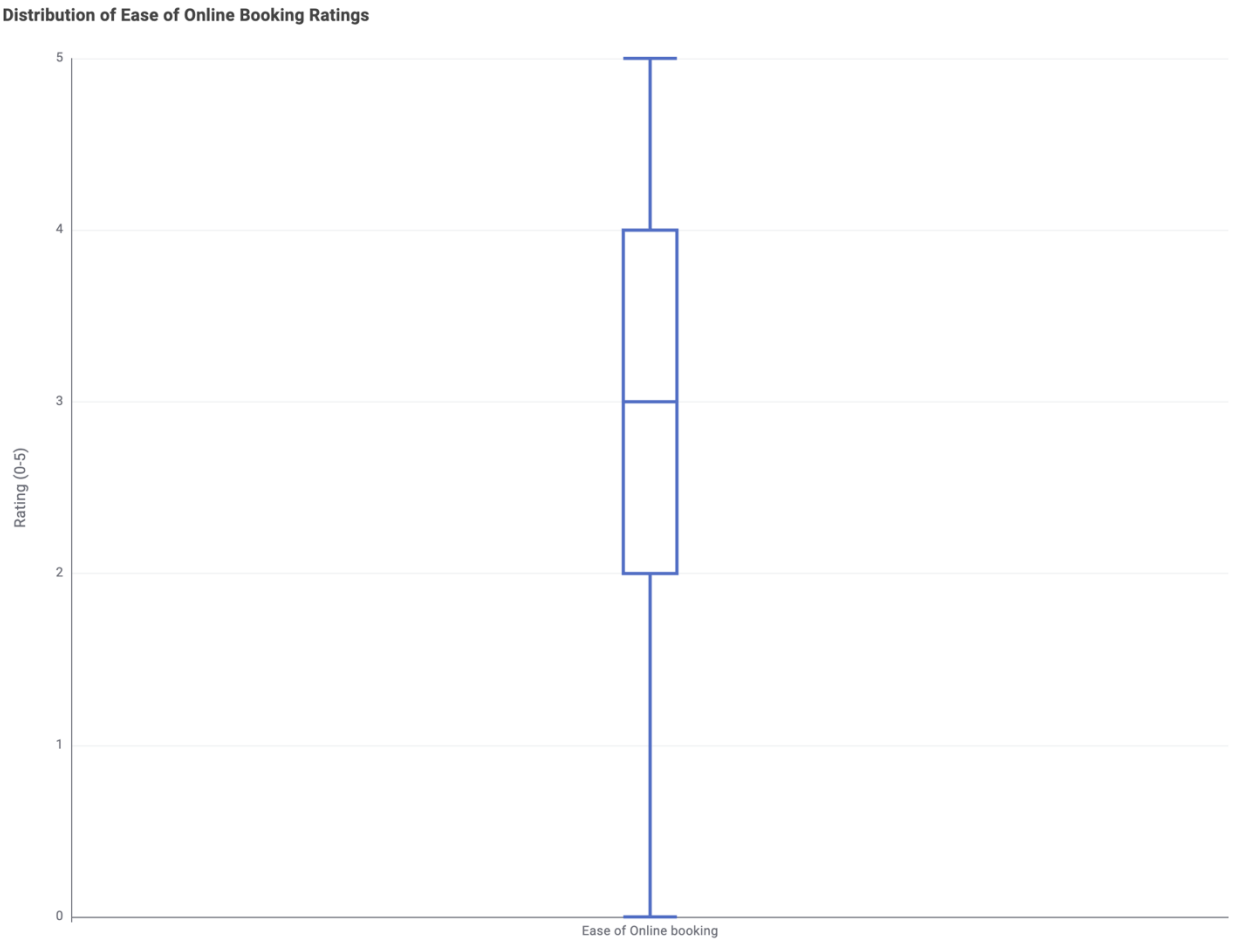


Figure 3: Boxplot of the *Distribution of Food and Drink Ratings (median = 3, IQR = 2-4, Scale = 0-5)*

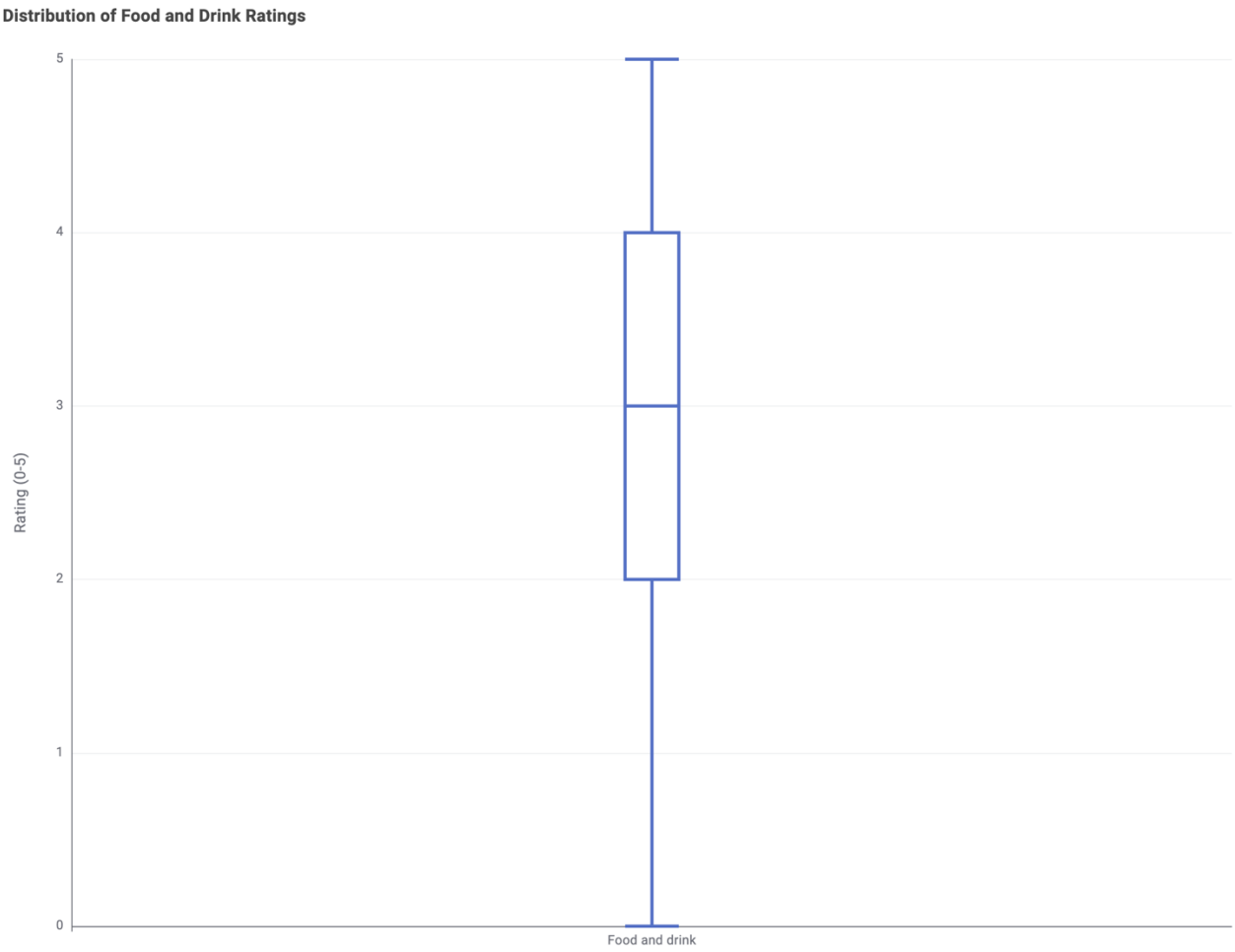
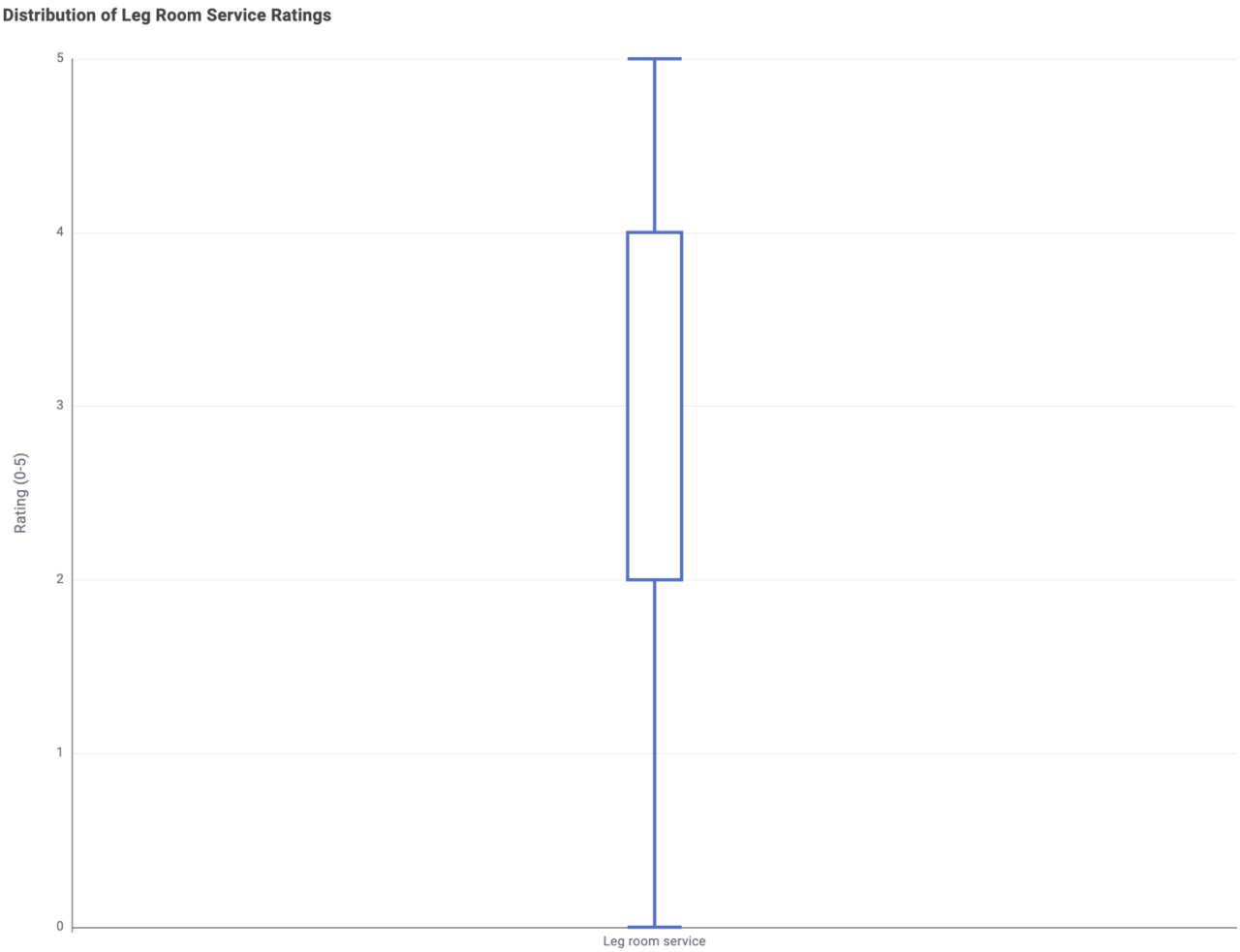


Figure 4: Boxplot of the *Distribution of Leg Room Service Ratings (median = 4, IQR = 2-4, Scale = 0-5)*



4.2.2 Boxplots of the Continuous Variables:

Boxplots identified 580 observations with Age = 999, which is not physically possible; these observations were recoded as missing values. Departure and arrival delay exhibited significant positive skewness (5.22 and 5.24, respectively) with numerous outliers reflecting genuine long delays. No transformations were applied, as GBT is unaffected by scale (Pinheiro et al., 2025), and LR's skewed variables are addressed through z-score standardisation in preprocessing.

Figure 5: *Distribution of Flight Distance (Distance measured in miles, minimum value = 56, median = 825, IQR = 406-1662, skewness = 1.16, maximum value = 4,983. Outliers are consistent with long-haul flights and are retained for analysis)*

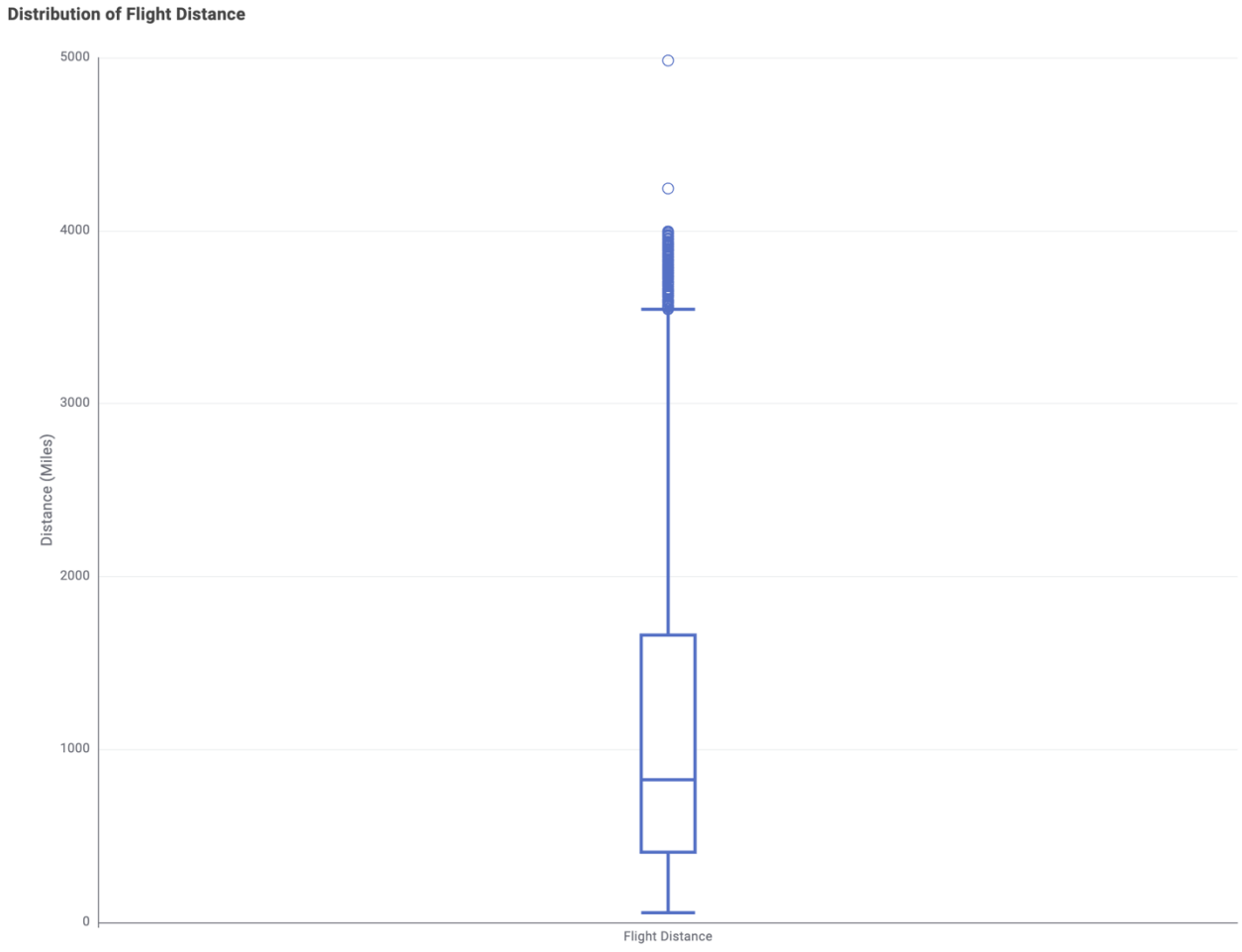


Figure 6: *Distribution of Departure Delay (Measured in minutes, minimum value = 0, median = 0, IQR = 0-12, skewness = 5.22, maximum value = 565. Outliers are consistent with actual departure delays and are retained for analysis)*

Note: *Departure delay was removed due to high multicollinearity with arrival delay. See section of this report.*

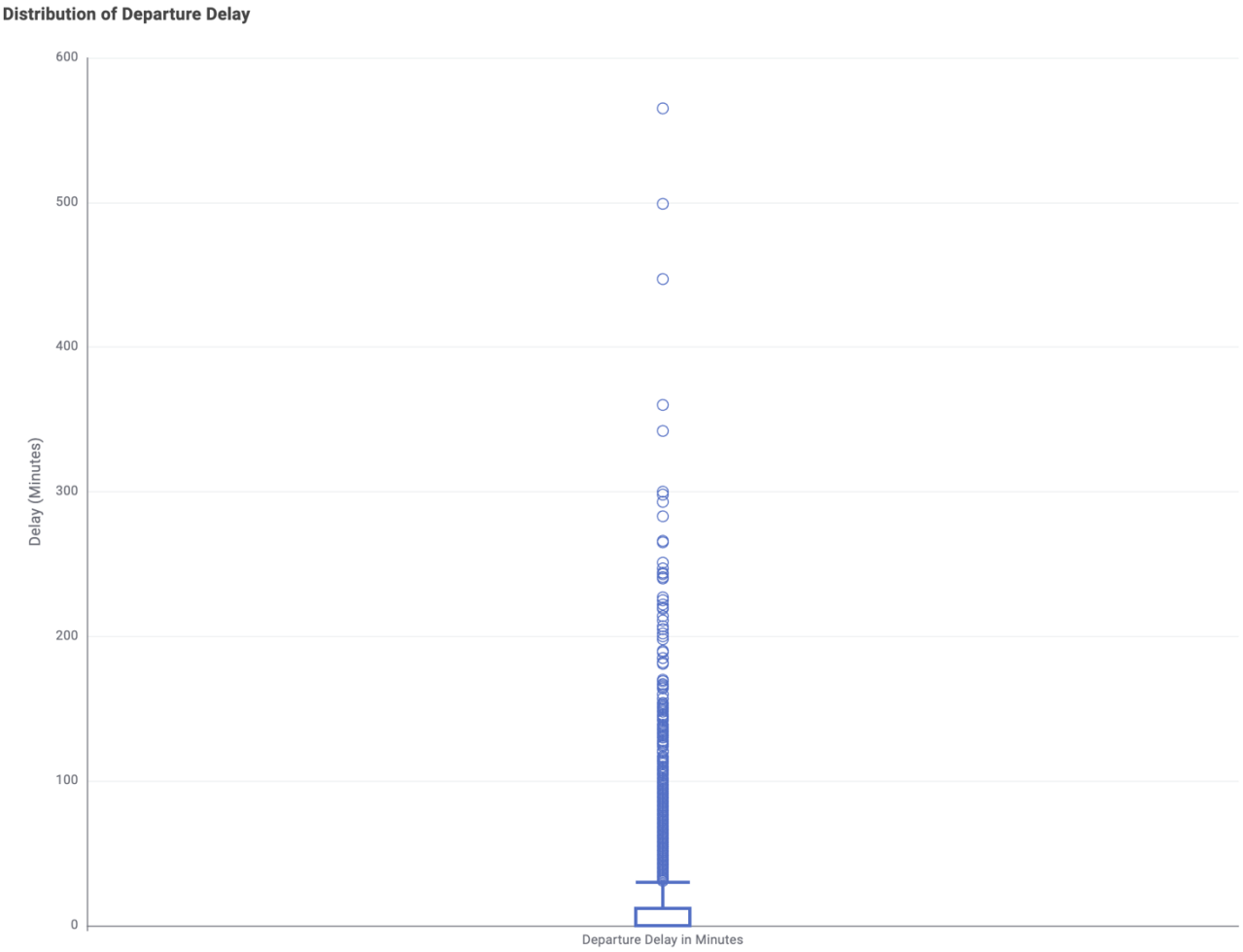


Figure 7: *Distribution of Arrival Delay (Measured in minutes, minimum value = 0, median = 0, IQR = 0-13, skewness = 5.24, maximum value = 586. Outliers are consistent with actual arrival delays and are retained for analysis)*

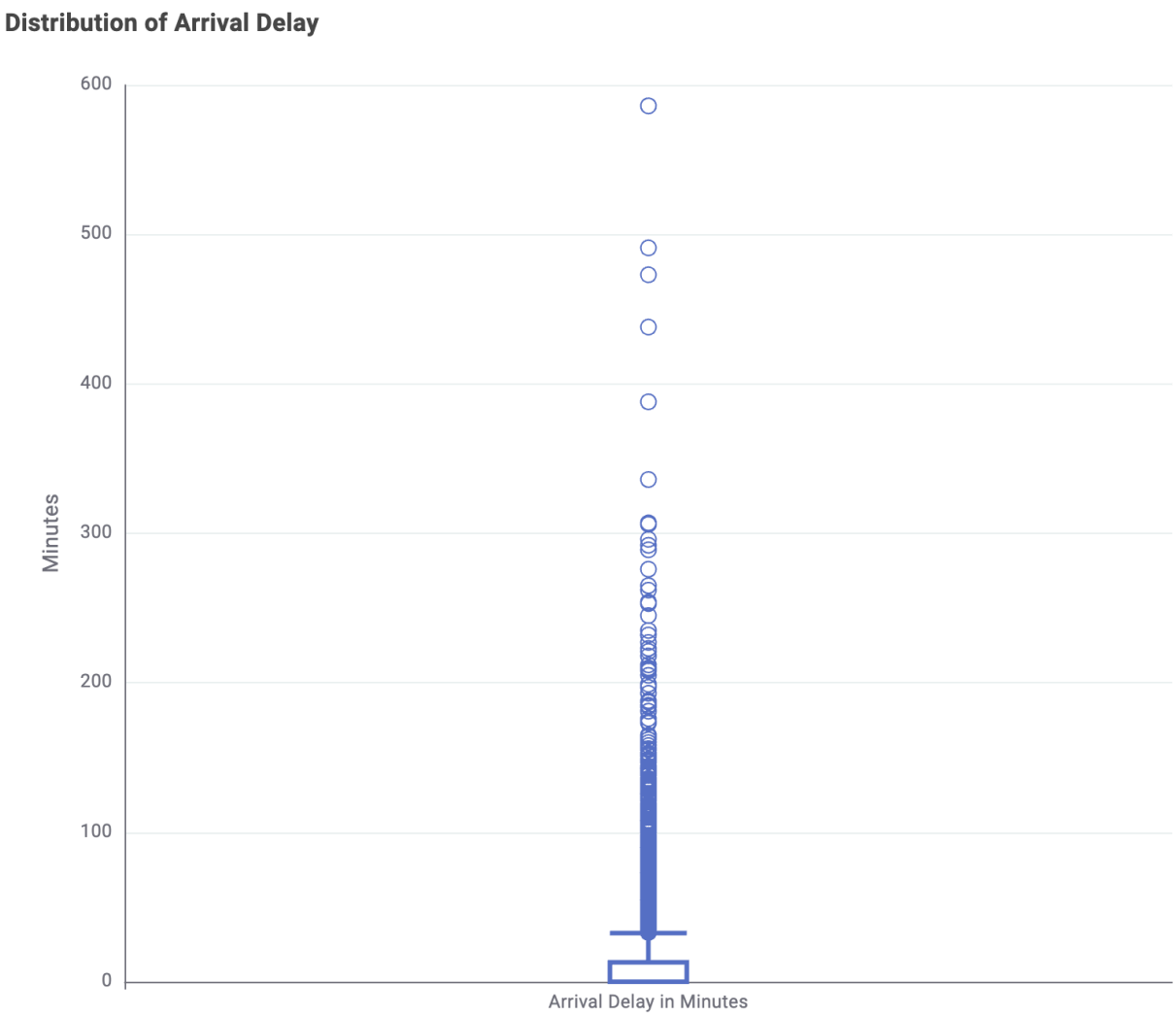
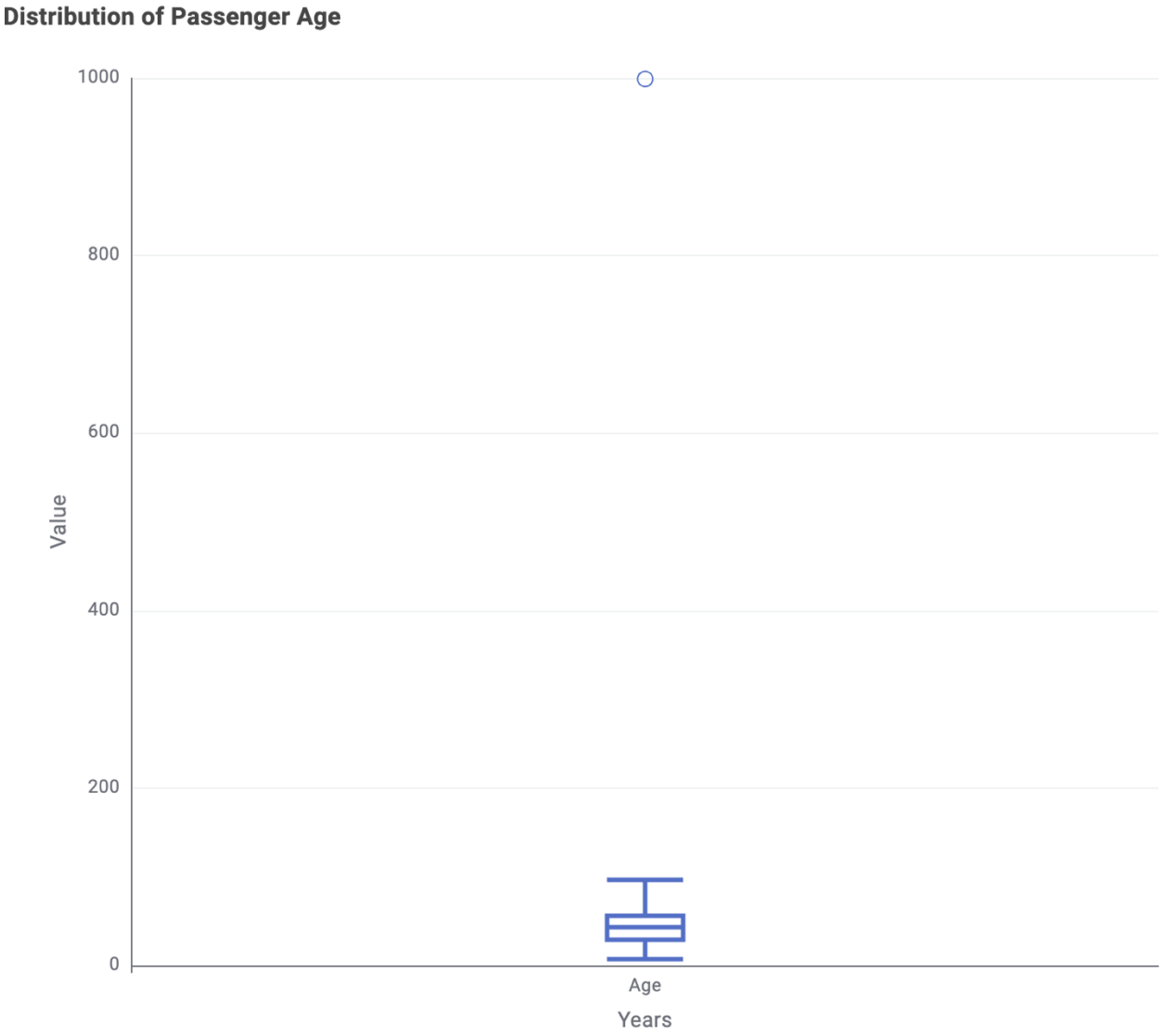


Figure 8: *Distribution of Age (Measured in years, minimum = 7, median = 43, IQR = 29-56, skewness = 2.46, maximum value = 999. Multiple outliers of age 999 which is not possible. Therefore, these were converted to missing values).*



4.2.3 Bar Charts of Nominal Categorical Variables:

Bar charts revealed that business class passengers, business travellers, cardholders, and passengers who checked in online were predominantly satisfied, while their counterparts were predominantly dissatisfied, suggesting strong associations with the dependent variable. The dataset was balanced for gender, with similar satisfaction distributions across both groups.

Figure 9: *Distribution of Passenger Satisfaction by Type of Travel (Business travel: 1,482 dissatisfied, 2,107 satisfied; Personal travel: 1,445 dissatisfied, 162 satisfied)*

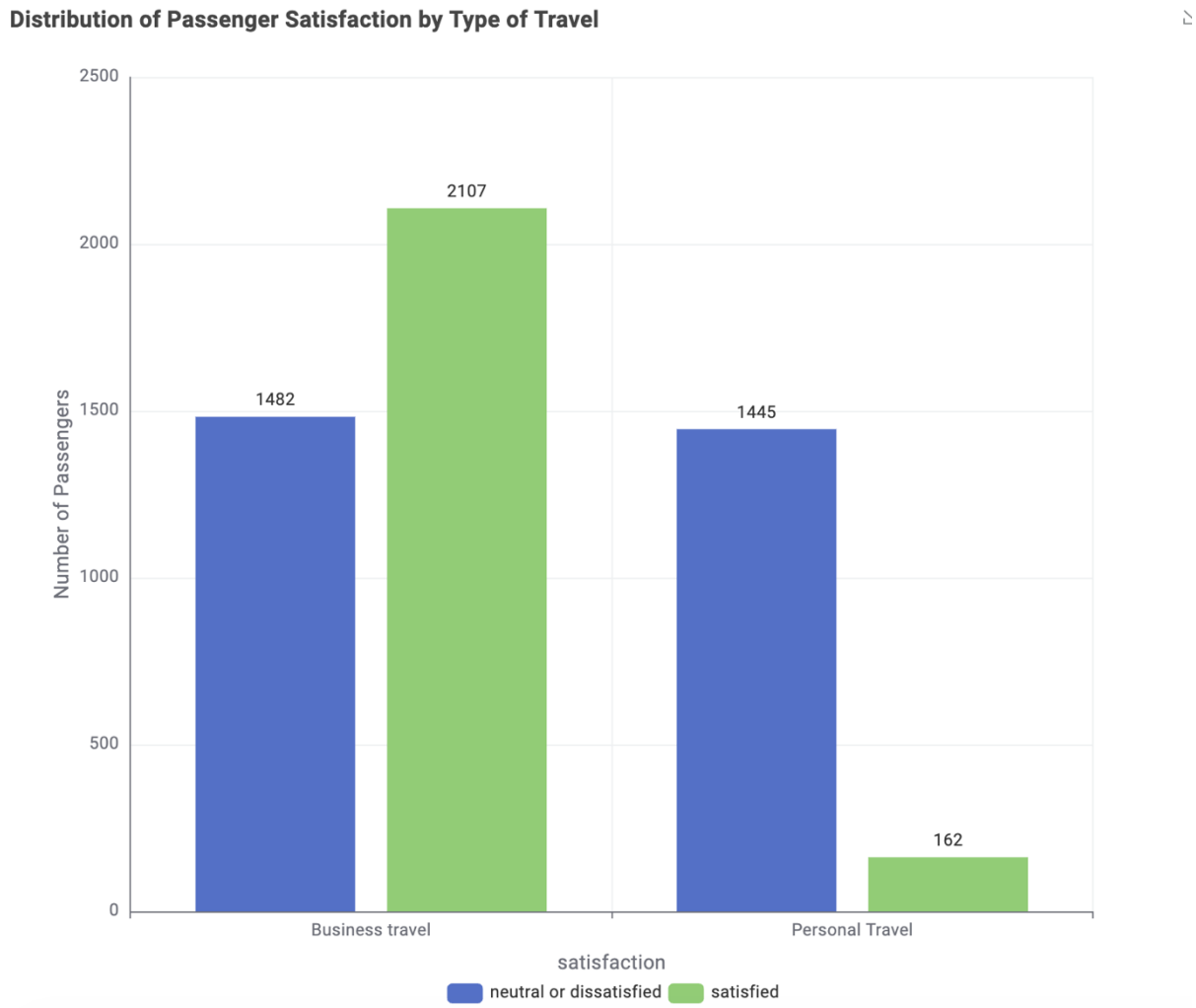


Figure 10: *Distribution of Passenger Satisfaction by Customer Loyalty Status (Cardholder: 2,216 dissatisfied, 2,016 satisfied; Non-cardholder: 711 dissatisfied, 253 satisfied)*

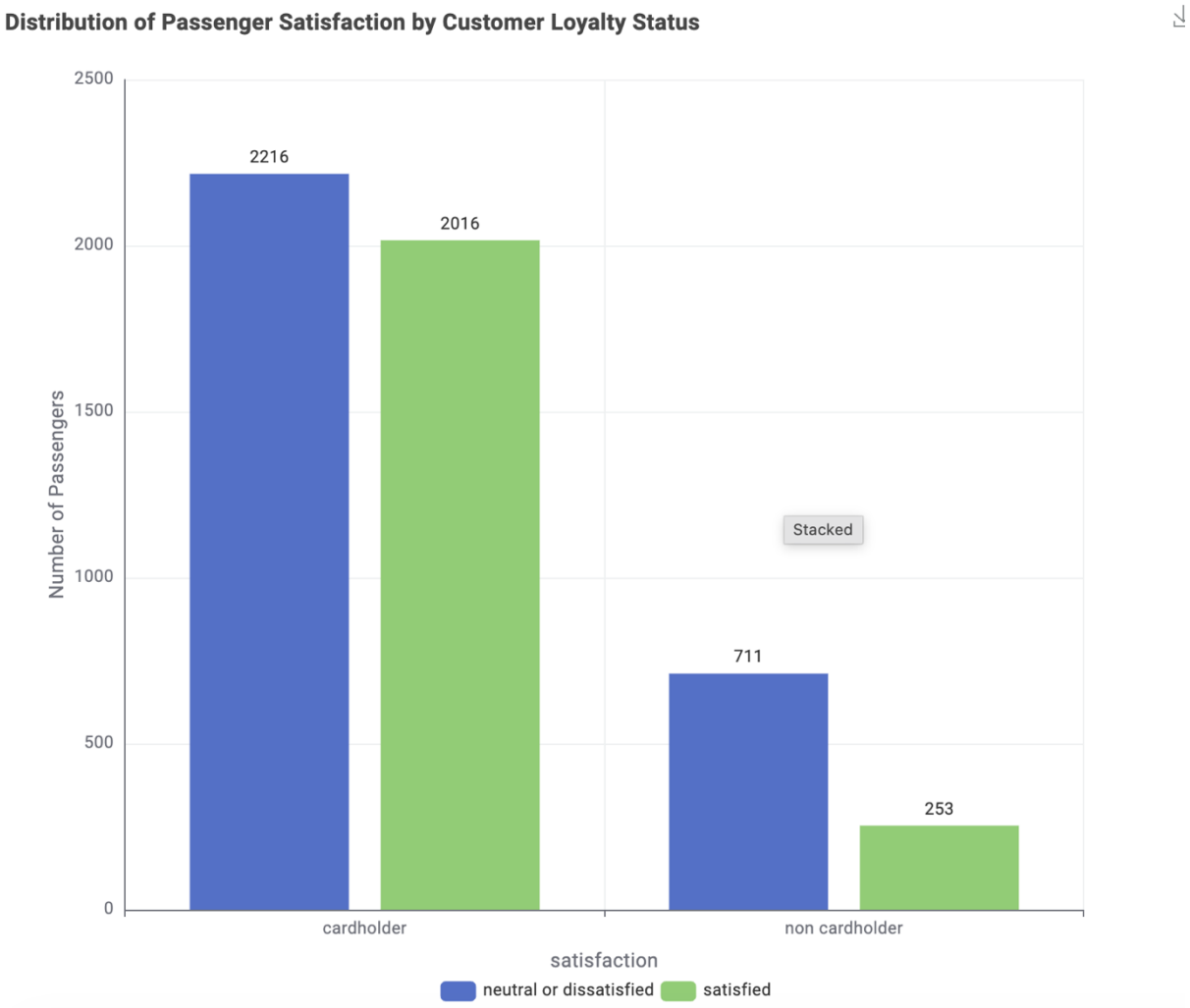


Figure 11: *Distribution of Passenger Satisfaction by Travel Class (Business: 799 dissatisfied, 1,691 satisfied; Economy: 1,863 dissatisfied, 470 satisfied; Economy Plus: 265 dissatisfied, 108 satisfied)*

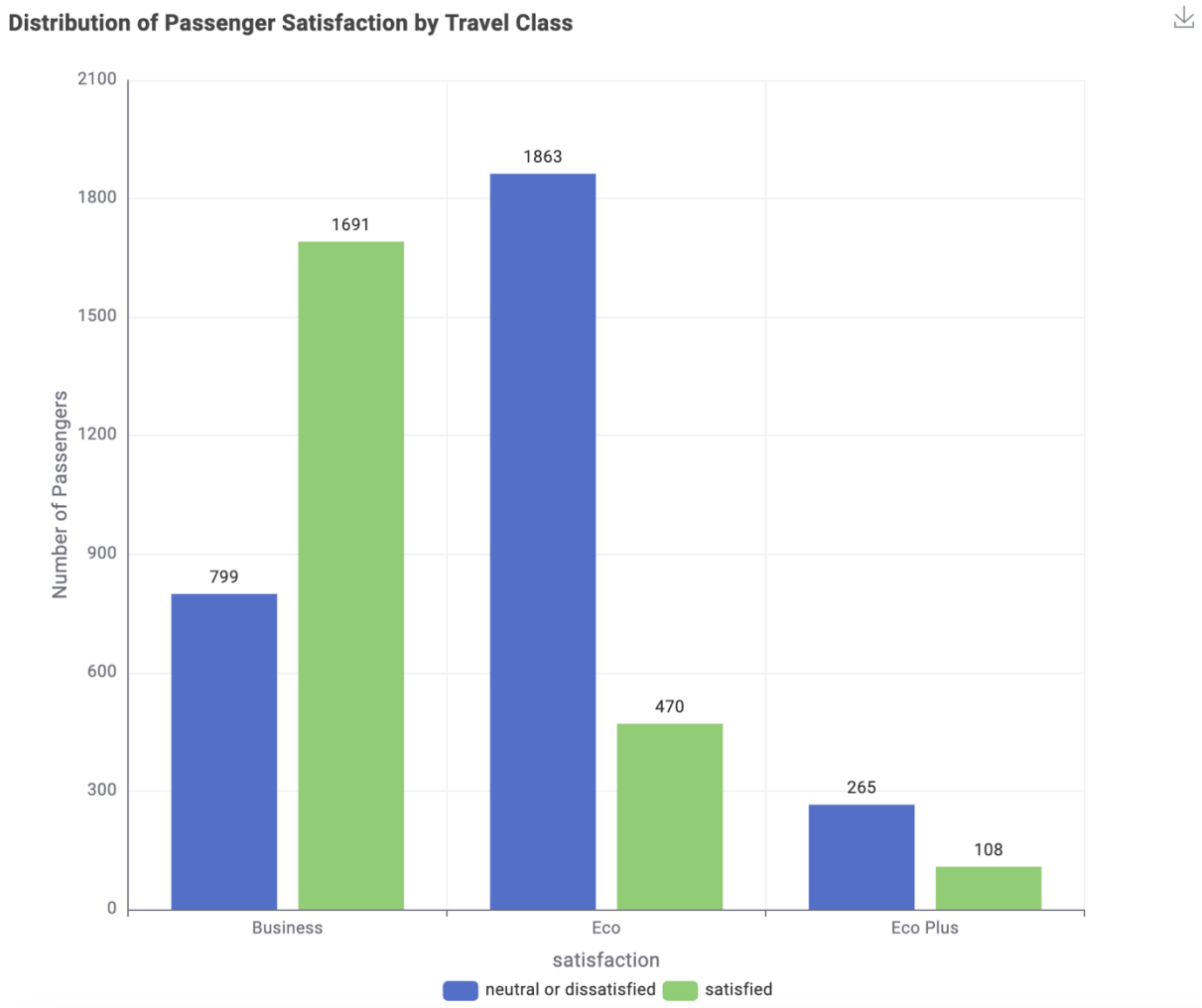
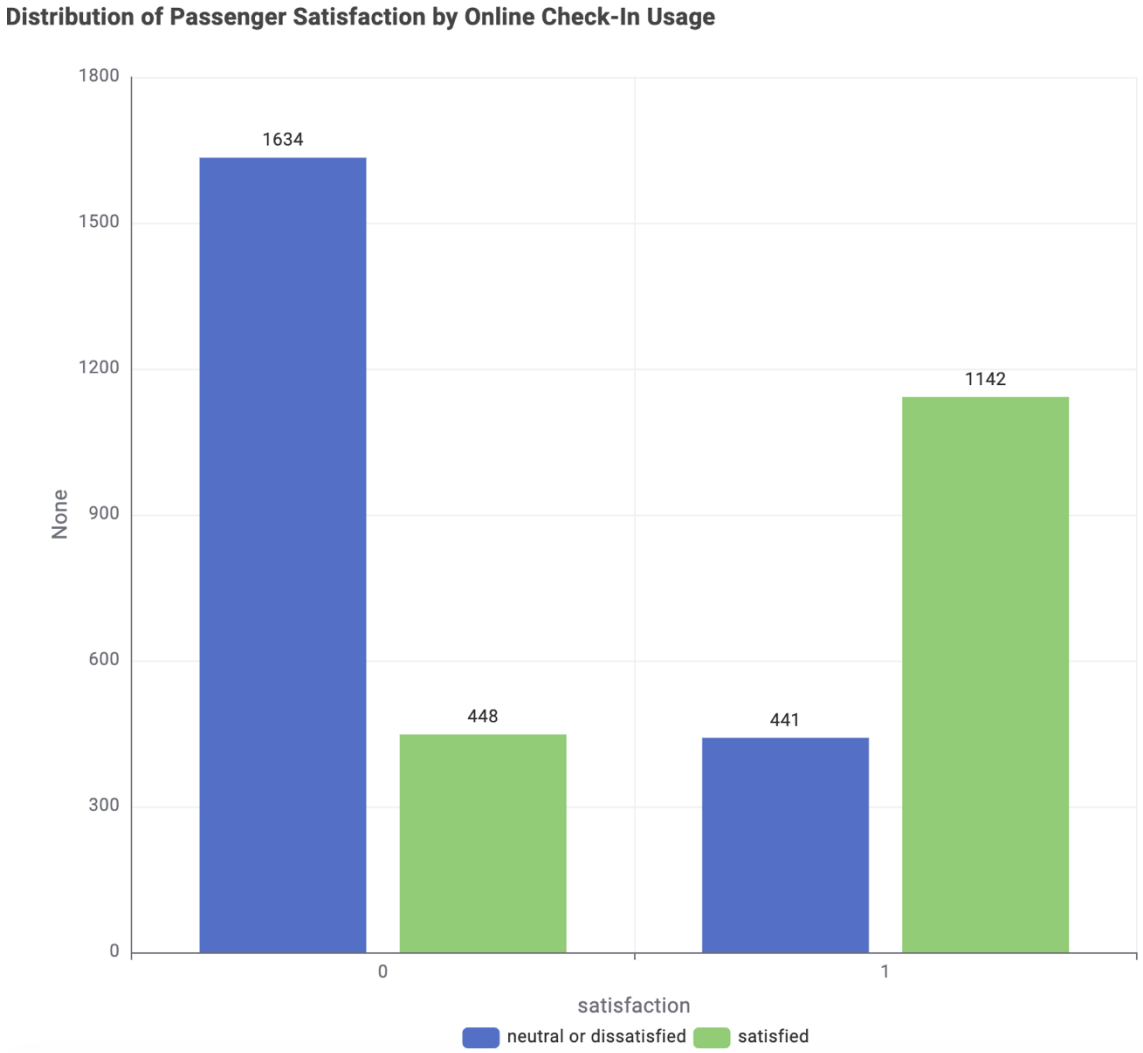


Figure 12: *Distribution of Passenger Satisfaction by Online Check-In Usage (No check-in: 1,634 dissatisfied, 448 satisfied; Online check-in: 441 dissatisfied, 1,142 satisfied)*



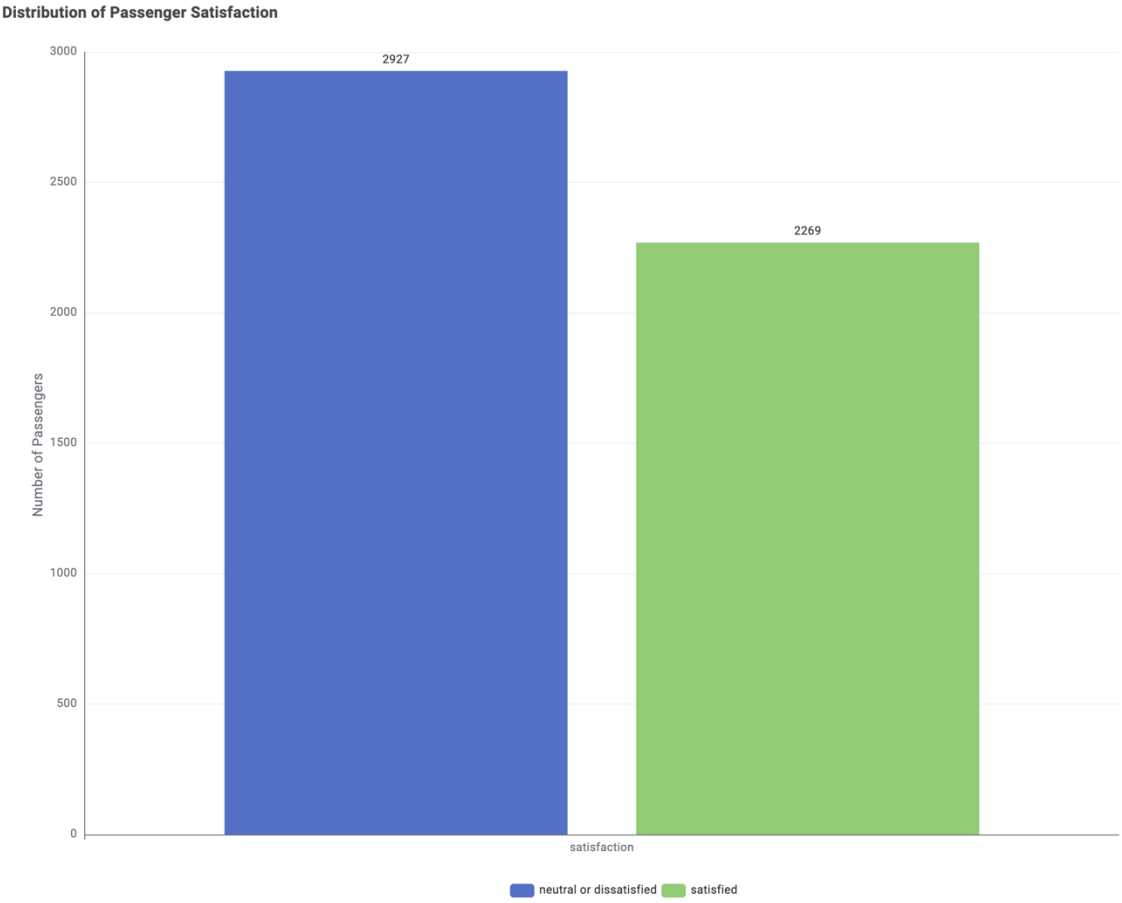
4.2.4 Dependent Variable Distribution:

Table 6 and Figure 13 show a mild class imbalance (neutral/dissatisfied = 56.3%, satisfied = 43.7%). SMOTE resampling was not applied as both models maintain reliable performance under mild imbalance (Sakho et al., 2026).

Table 6: *Class distribution of the dependent variable (Satisfaction).*

Class	Count (n)	Percentage (%)
Neutral or Dissatisfied	2,927	56.3%
Satisfied	2,269	43.7%
Total	**5,196**	**100%**

Figure 13: *Distribution of Passenger Satisfaction (Neutral/Dissatisfied = 2,927; Satisfied = 2,269)*



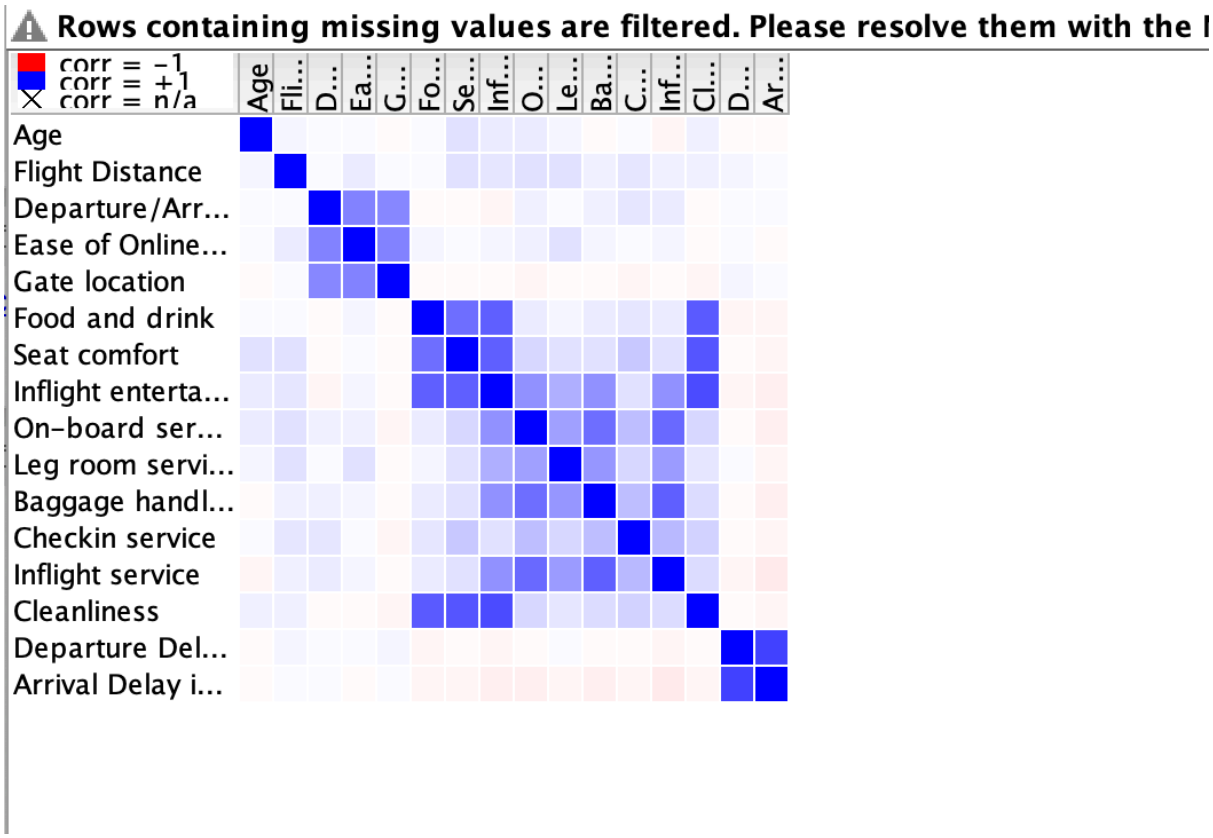
4.2.5 Correlation Analysis and Multicollinearity:

Spearman correlation was selected over Pearson due to the presence of ordinal and heavily skewed continuous predictor variables (Hauke & Kossowski, 2011). Table 7 and Figure 14 display the Spearman correlation matrix. The matrix revealed a strong positive correlation between departure delay and arrival delay ($\rho = 0.75$), exceeding the $|\rho| > 0.7$ threshold, suggesting high collinearity between these predictors (Dormann et al., 2012; Harrison et al., 2018). Departure delay was therefore removed from the logistic regression pipeline after partitioning to prevent multicollinearity from increasing the standard errors and making the coefficients harder to interpret (Luqman et al., 2025). Arrival delay was retained, as it had significantly fewer missing values (15 versus 747). GBT retained both variables as it is robust to correlated predictors through built-in feature selection (Li et al., 2022).

Table 7: *Spearman rank correlation matrix for continuous and ordinal predictor variables. Highlighted cell indicates high correlation between Departure Delay and Arrival Delay (ρ = 0.75).*

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Age	**1.00**	0.04	0.02	0.01	0.00	0.02	0.12	0.07	0.06	0.03	-0.01	0.02	-0.04	0.04	-0.01	-0.01
2. Flt Dist		**1.00**	0.01	0.07	0.01	0.02	0.12	0.09	0.10	0.11	0.06	0.08	0.06	0.06	0.03	0.00
3. Dep/Arr Time			**1.00**	0.47	0.47	-0.01	0.00	-0.03	0.05	0.02	0.05	0.10	0.08	-0.02	0.01	0.01
4. Online Book				**1.00**	0.48	0.03	0.01	0.03	0.05	0.11	0.04	0.02	0.04	0.00	0.00	-0.01
5. Gate Loc					**1.00**	-0.01	0.00	-0.02	-0.02	-0.01	-0.02	-0.04	-0.02	-0.03	0.02	0.02
6. Food & Drink						**1.00**	0.56	0.62	0.07	0.03	0.07	0.08	0.06	0.64	-0.03	-0.04
7. Seat Comfort							**1.00**	0.61	0.14	0.10	0.12	0.20	0.10	0.66	-0.02	-0.04
8. Inflight Ent								**1.00**	0.43	0.31	0.42	0.12	0.43	0.70	-0.02	-0.05
9. On-board									**1.00**	0.37	0.55	0.25	0.57	0.14	-0.02	-0.05
10. Leg Room										**1.00**	0.41	0.15	0.39	0.09	0.01	-0.02
11. Baggage											**1.00**	0.25	0.63	0.14	-0.02	-0.05
12. Checkin												**1.00**	0.27	0.17	-0.01	-0.04
13. Inflight Svc													**1.00**	0.13	-0.03	-0.07
14. Cleanliness														**1.00**	-0.02	-0.03
15. Dep Delay															**1.00**	**0.75**
16. Arr Delay																**1.00**

Figure 14*: Spearman rank correlation heatmap from KNIME for the continuous and ordinal predictor variables. The strong correlation between Departure Delay and Arrival Delay (ρ = 0.75) is visible in the bottom-right cluster*



4.3 Data Preparation:

4.3.1 Train/Test Partitioning:

The dataset was split 80/20 into training (n = 4,156) and test (n = 1,040) sets before preprocessing, using stratified sampling to preserve class proportions and prevent data leakage. An 80/20 ratio was selected to maximise training sample size while retaining a sufficient test set. Departure delay was removed from the LR pipeline after partitioning.

4.3.2 Missing Values and Imputation:

GBT handles missing values automatically (Chen & Guestrin, 2016), so imputation was applied only to the LR pipeline. Listwise deletion was avoided as it only produces unbiased estimates under MCAR (Peng et al., 2023). Median imputation was applied to continuous variables due to skewness and extreme outliers (Alam et al., 2023), and mode

imputation to categorical variables as standard practice (Ge et al., 2023; Jager et al., 2021). Table 8 summarises the missingness classification and treatment for each variable.

Table 8: *Missingness Classification and Treatment (Imputation only applied to the LR model).*

Variable	Missing %	Missing (n)	Possible Mechanism	Mechanism Justification	Treatment
Flight Distance	14.4%	747	Could be MCAR or MAR	Could be MCAR. Also, all 747 affected rows have service ratings confirming valid flights. Therefore, missingness could be related to airline or route but not related to the variable itself (MAR) (Little & Rubin, 2003)	Median imputation
Arrival Delay	0.29%	15	Could be MCAR or MAR	All missing rows have service ratings confirming valid flights. Same reasons as flight distance above.	Median imputation
Age (999s)	11.2%	580	Could be MCAR or MNAR	Could reflect random data entry errors (MCAR) or deliberate non-disclosure by older passengers (MNAR)	Median imputation
Online Check-in	29.5%	1,531	Could be MCAR or MNAR	Could be MCAR or older people who aren't as tech savvy, so they didn't check in online (MNAR)	Mode imputation
Departure/Arrival Time Convenience	5.3%	277	MCAR or MNAR	Documented as 'not applicable' even though departure/arrival time convenience	Mode imputation

Variable	Missing %	Missing (n)	Possible Mechanism	Mechanism Justification	Treatment
				would apply to all passengers. Could be MCAR. Alternatively, dissatisfied passengers may be more likely to avoid rating services, consistent with known positivity bias in customer satisfaction surveys where dissatisfied customers are less likely to respond (Park et al., 2018)	
Ease of Online Booking	4.4%	229	MCAR or MNAR	Same justification as Departure/Arrival Time Convenience above.	Mode imputation
Food and Drink	0.10%	5	MCAR or MNAR	Same justification as Departure/Arrival Time Convenience above.	Mode imputation
Leg Room Service	0.54%	28	MCAR or MNAR	Same justification as Departure/Arrival Time Convenience above.	Mode imputation

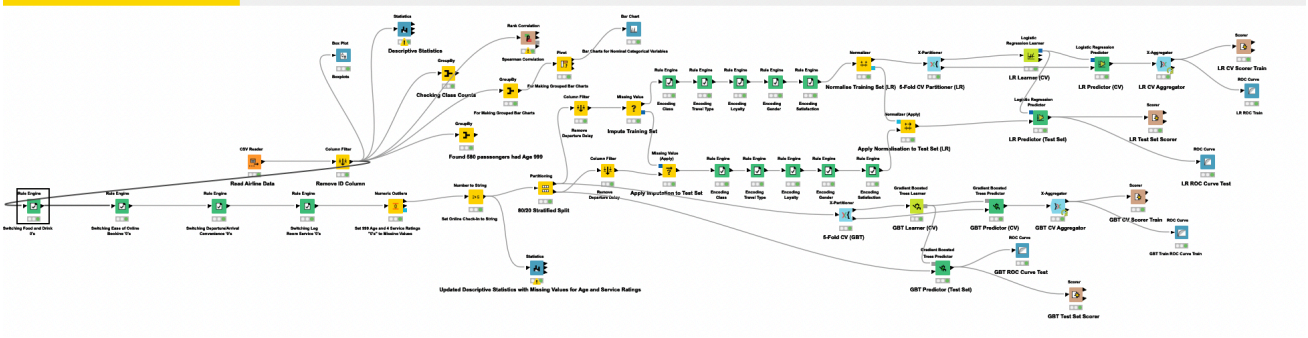
4.3.3 Encoding and Normalisation:

Nominal categorical variables were manually encoded for LR with reference categories chosen for interpretability. Z-score standardisation was applied to continuous variables in the LR pipeline as coefficient magnitudes depend critically on input scales (Arafat et al., 2025). Z-score normalisation was selected over min-max normalisation due to flight distance and arrival delay having extreme but valid outliers. As noted by Laska and Yolanda (2024), min-max normalisation is sensitive to outliers as it scales data to a fixed

range, whereas z-score normalisation is resistant to outliers as it standardises data using the mean and standard deviation without compressing extreme values.

5. Experimental Setup

Figure 15: *Overview of KNIME Workflow*



5.1 Cross-Validation:

5-fold cross-validation was applied to the training set for both models. Cross-validation reduces the risk of overfitting to the training data (Abedin et al., 2026). It also produces more stable performance estimates than a single train-test split (Singh et al., 2021). This also enables a more reliable comparison of hyperparameter configurations across models. 5-fold was chosen over 10-fold or higher, as increasing the number of folds increases the computational cost (Oyedele, 2023).

5.2 Hyperparameter Tuning:

Hyperparameter tuning was conducted for both models to identify the optimal configuration, as hyperparameters critically affect a model's ability to generalise to unseen data and default settings are not guaranteed to provide the optimal predictive performance for this dataset (Arnold et al., 2024).

5.2.1 Logistic Regression Hyperparameter Configurations:

All regularisation types were initially tested at 100 epochs, none of which converged. Epochs were increased to 500. Phase 1 tested Gaussian (L2) regularisation across five variance values. Only variance = 0.1 converged, indicating strong L2 regularisation was required. Phase 2 confirmed Gaussian (L2) as the appropriate regularisation type, as both

Laplace (L1) and Uniform failed to converge at variance = 0.1 and 500 epochs. The optimal configuration was Gaussian (L2), variance = 0.1, and 500 epochs.

Table 9: *LR Hyperparameter Configurations (highlighted had the best metrics and was selected. The other configurations failed to converge)*

Regularisation	Variance	Epochs	F1	AUC-ROC	Cohen's Kappa	Accuracy
Uniform (No Regularisation)	0.1	100	—	—	—	—
Gauss (L2)	0.1	100	—	—	—	—
Laplace (L1)	0.1	100	—	—	—	—
Gauss (L2)	**0.1**	**500**	**.83**	**.92**	**.71**	**.86**
Gauss (L2)	0.5	500	—	—	—	—
Gauss (L2)	1.0	500	—	—	—	—
Gauss (L2)	5.0	500	—	—	—	—
Gauss (L2)	10.0	500	—	—	—	—
Laplace (L1)	.1	500	—	—	—	—
Uniform (No Regularisation)	.1	500	—	—	—	—

5.2.2 Gradient Boosted Trees Hyperparameter Configurations:

Three hyperparameters were tuned sequentially. Phase 1 fixed trees (100) and depth (4), identifying learning rate = 0.1 as optimal. Phase 2 fixed learning rate (0.1) and trees (100), identifying depth = 6 as optimal. Phase 3 fixed learning rate (0.1) and depth (6), confirming 100 trees as optimal. The selected configuration was learning rate = 0.1, 100 trees, and maximum depth = 6.

Table 10: *GBT Hyperparameter Configurations (highlighted had the best metrics and was selected)*

Learning Rate	Trees	Depth	F1	AUC-ROC	Cohen's Kappa	Accuracy
0.2	100	4	.90	.97	.82	.91
0.1	100	4	.90	.97	.83	.92
0.05	100	4	.90	.97	.83	.91

0.01	100	4	.88	.96	.79	.90
0.1	100	3	.90	.97	.83	.92
0.1	**100**	**6**	**.91**	**.97**	**.84**	**.92**
0.1	100	8	.90	.97	.83	.92
0.1	50	6	.90	.97	.83	.92
0.1	200	6	.91	.97	.83	.92

5.3 Key KNIME Nodes for Experimental Setup:

Table 11 presents the most crucial KNIME nodes used in the experimental setup. Nodes used for exploratory data analysis and basic data filtering were not included in the table.

Table 11: *Key KNIME Nodes and Justification for Experimental Setup*

KNIME Nodes	Pipeline	Justification
Partitioning	Both	• Split data 80/20 into training and test sets using stratified sampling to preserve class proportions. • Split was conducted before pre-processing to prevent data leakage, which leads to biased and inflated performance metrics (Kapoor & Narayanan, 2023). • An 80/20 ratio was selected to maximise training sample size while retaining a sufficient test set for reliable performance evaluation. • Set seed for reproducibility.
Column Filter	LR only	• Removed Departure Delay due to high collinearity with Arrival Delay ($\rho = 0.75$). • LR models are sensitive to multicollinearity, which can increase standard errors and make coefficients harder to interpret (Luqman et al., 2025).
Missing Value	LR only	• LR cannot handle missing values, as it requires complete data on the predictor variables in order to calculate their coefficients (Das et al., 2010). • Applied median imputation for the continuous variables and mode for Online Check-in and the four service rating variables.

		• Median imputation was applied to all continuous variables as it is more robust to the skewness and extreme outliers present in Flight Distance and Arrival Delay (Alam et al., 2023). Although mean imputation would have been more appropriate for Age given its near-symmetrical distribution (skewness = -0.001 after 999 were switched to missing values), the difference between mean and median imputation values was negligible (39.78 vs 40.00 after 999 were switched to missing values), confirming minimal impact on model performance. Mode imputation was applied to the online check-in and service rating variables with missing values, as this is standard practice for categorical variables (Ge et al., 2023; Jager et al., 2021).
Rule Engine	LR only	• Used Rule Engine nodes to encode the nominal categorical variables and choose the appropriate reference category for each of them.
Normalizer (Training Set) and Normalizer (Apply) (Test Set)	LR only	• Logistic Regression requires careful feature standardisation as coefficient magnitudes depend critically on input scales (Arafat et al., 2025), meaning continuous predictor variables on different scales produce incomparable coefficients and cause gradient-based optimisers to converge poorly. • Z-score normalisation was selected over min-max normalisation due to flight distance and arrival delay having extreme but valid outliers. As noted by Laska and Yolanda (2024), min-max normalisation is sensitive to outliers as it scales data to a fixed range, whereas z-score normalisation is resistant to outliers as it standardises data using the mean and standard deviation without compressing extreme values.
Cross-Validation Loop (X-Partitioner and X-Aggregator Nodes)	Both	• 5-fold cross-validation was applied to the training set for both models. Cross-validation partitions the training data into k folds, training on k-1 folds and validating on the remaining fold, repeating k times, thereby reducing the risk of overfitting to the training data (Abedin et al., 2026).

		• Cross-validation was conducted on both models because it reduces the risk of overfitting to the training data (Abedin et al., 2026). It also produces more stable performance estimates than a single train-test split (Singh et al., 2021). • 5-fold was chosen over 10-fold or higher, as increasing the number of folds increases the computational cost (Oyedele, 2023). • Set seed for reproducibility.
Logistic Regression Learner	LR only	• Hyperparameter tuning was conducted for both models to identify the optimal configuration, as hyperparameters critically affect a model's ability to generalise to unseen data and default settings are not guaranteed to provide the optimal predictive performance for this dataset (Arnold et al., 2024). • Logistic regression hyperparameter tuning focused on regularisation type and variance. Regularisation controls model complexity and prevents overfitting by penalising large coefficients, making the selection of the appropriate regularisation strength critical for model generalisation (Greenwood et al., 2020). • Selected Stochastic Average Gradient Optimiser, with Gaussian (L2) regularisation, variance = 0.1, and 500 epochs. • Set seed for reproducibility.
GBT Learner	GBT only	• Gradient boosted trees hyperparameter tuning varied three key hyperparameters. Learning rate controls each tree's contribution to the ensemble, with too low a rate requiring more trees to compensate, thus risking underfitting (Natekin & Knoll, 2013). Max depth controls tree complexity, with deeper trees capturing more patterns but increasing the risk of overfitting (Maniwa et al., 2024). Number of trees determines ensemble size, with too few trees leading to underfitting and too many risking overfitting (Natekin & Knoll, 2013). • Selected learning rate = 0.1, 100 trees, and max depth = 6.

		• Set seed for reproducibility.
Scorer	Both	• Calculated accuracy, F1-score, Cohen's Kappa, and confusion matrices for both test and training sets.
ROC Curve	Both	• Generated ROC curves and AUC-ROC for both test and training sets.

6. Results and Discussion

6.1 Model Performance:

Table 12 presents the four performance metrics on the training and test sets for both models. The minimal reduction across all metrics from training to test set indicates that neither model was overfitted on the training data, suggesting that 5-fold cross-validation assisted in preventing overfitting (Abedin et al., 2026). Gradient boosted trees outperformed the logistic regression model on all metrics across both the training and test sets. GBT's higher AUC-ROC (0.97 vs 0.91) confirms superior class separation across all thresholds, indicating that GBT more reliably distinguishes satisfied from dissatisfied passengers regardless of the decision boundary (Li, 2024). Its higher F1-score (0.89 vs 0.82) indicates stronger performance across both satisfaction classes, meaning GBT missed fewer satisfied passengers and had fewer false positive predictions (Tharwat, 2021). GBT's Cohen's Kappa of 0.80 versus LR's moderate 0.68 suggests that a notable proportion of LR's correct predictions are attributable to chance rather than its genuine discriminative ability (McHugh, 2012). GBT also achieved higher accuracy (0.90 vs 0.84), though given the mild class imbalance (Tharwat, 2021), AUC-ROC, F1-score, and Cohen's Kappa are more informative measures of performance.

Table 12: *Performance Metrics for Both Models (highlighted indicates the best metrics and model).*

Model	Set	Accuracy	F1-Score	AUC-ROC	Cohen's Kappa
LR	Training (CV)	0.86	0.83	0.92	0.71
LR	Test	0.84	0.82	0.91	0.68

GBT	Training (CV)	0.92	0.91	0.97	0.84
GBT	**Test**	**0.90**	**0.89**	**0.97**	**0.80**

Figure 16: *ROC Curve of Logistic Regression on Test Set*

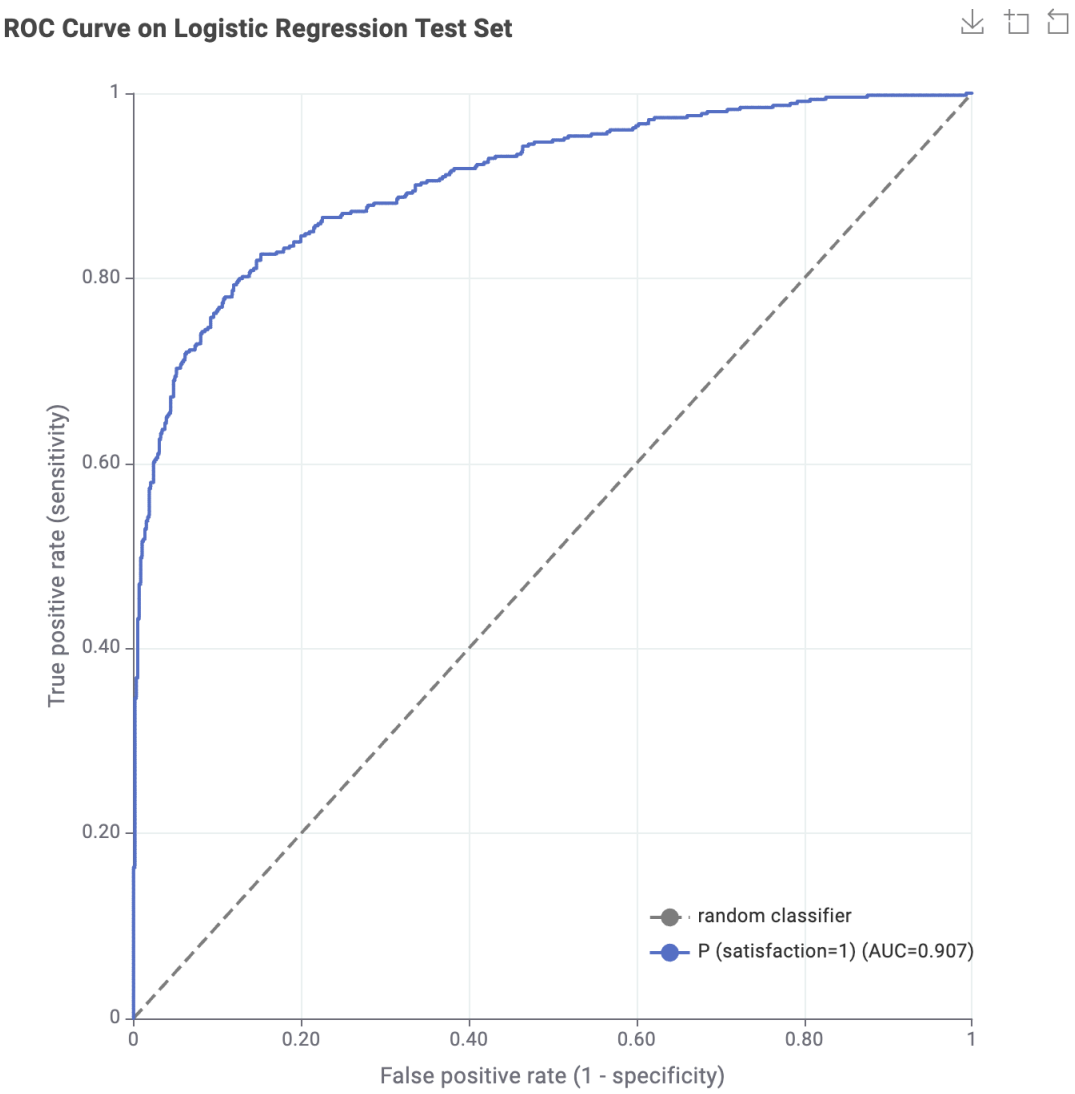
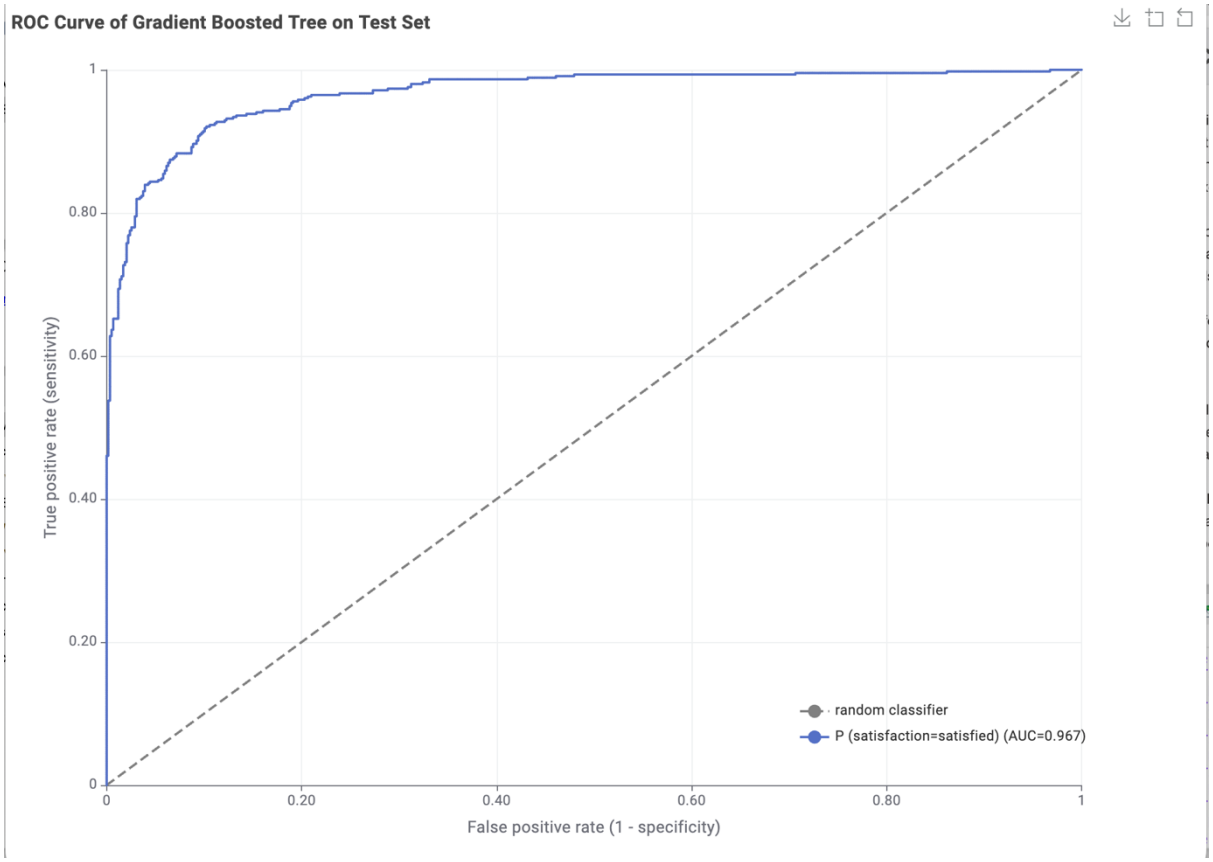


Figure 17: *ROC Curve of Gradient Boosted Trees on Test Set*



6.2 Confusion Matrix Analysis:

Tables 13 and 14 present the test set confusion matrices for both models. GBT reduced false negatives by 28 (64 vs 92) and false positives by 36 (36 vs 72). Correctly identifying satisfied passengers enables airlines to employ targeted loyalty and retention strategies to the right customers, while avoiding false satisfaction predictions prevents costly misallocation of service improvement resources towards passengers who are not genuinely satisfied. This is particularly relevant given that airlines are still financially recovering from the pandemic.

Table 13: *Logistic Regression Confusion Matrix on the Test Set*

	Pred: Neutral/Dissatisfied	Pred: Satisfied
Actual: Neutral/Dissatisfied	514 (True Negatives)	72 (False Positives)
Actual: Satisfied	92 (False Negatives)	362 (True Positives)

Table 14: *Gradient Boosted Trees Confusion Matrix on the Test Set*

	Pred: Neutral/Dissatisfied	Pred: Satisfied
Actual: Neutral/Dissatisfied	550 (True Negatives)	36 (False Positives)
Actual: Satisfied	64 (False Negatives)	390 (True Positives)

6.3 Why Gradient Boosted Trees Outperformed Logistic Regression:

GBT's superior performance is consistent with prior literature showing tree ensemble methods consistently outperform LR on airline satisfaction datasets (Bilgin, 2025; Jiang et al., 2022). The three dataset-specific factors identified in section 3.2, LR's linearity assumption, sensitivity to multicollinearity, and reliance on imputation, likely contributed to the performance gap observed.

6.4 Insights from Logistic Regression Coefficients:

As feature importance is unavailable in KNIME 5.3.3, the LR coefficient analysis below partially addresses this limitation. Table 15 presents the six most influential LR predictors. Travel type shows the largest coefficient in the model ($\beta = +2.339$, OR = 10.37): business travellers have 937% higher odds of satisfaction than personal travellers. Class was also a significant predictor, with Business class passengers being 108.1% more likely to be satisfied than passengers in economy class, both consistent with Jiang et al. (2022). This likely reflects that business travellers disproportionately occupy business class, which has more comfort, amenities, and services compared to economy class. Cardholders have 352% higher odds of satisfaction than non-cardholders, and may reflect the enhanced service access and benefits, such as priority boarding, class upgrades and lounge access, that loyalty

programmes provide. Passengers who checked in online were 165.6% more likely to be satisfied, and each additional rating point in the ease of online booking rating scale raises satisfaction odds by 78.6%, reinforcing the importance of digital service quality (Jiang et al., 2022). Regarding negative predictors, each additional standard deviation increase in arrival delay reduces satisfaction odds by 30.4% ($\beta = -0.362$). Gate Location's counterintuitive negative coefficient ($\beta = -0.381$) likely reflects multicollinearity distortion (Luqman et al., 2025).

Table 15: *Six Most Influential LR Predictors ($p < .05$)*

Rank	Variable	β	OR	P>|z|	Interpretation
1	Type of Travel=2 (Business) Reference Category: personal travellers	+2.339	10.37	< .001	Business travellers have 937% higher odds of satisfaction than personal travellers (reference)
2	Customer Loyalty=2 (Cardholder) Reference: Non-cardholder	+1.508	4.52	< .001	Cardholders have 352% higher odds of satisfaction than non-cardholders (reference)
3	Online Check-in=1 (Used) Reference: Didn't use	+0.977	2.656	< .001	Passengers who used online check-in have 165.6% higher odds than those who did not (reference)
4	Class=3 (Business Class) Reference: Economy	+0.733	2.081	< .001	Business passengers have 108.1% higher odds than economy passengers (reference)
5	Ease of Online Booking	+0.580	1.786	< .001	Each additional rating point raises odds of satisfaction by 78.6%
6	Arrival Delay	-0.362	0.696	< .001	Each additional standard deviation increase in arrival delay reduces odds of satisfaction by 30.4%

6.5 Recommendations for Airline Management:

These findings offer actionable insights for airline management. Airlines should prioritise loyalty programme growth through targeted incentives, improve digital service touchpoints particularly online check-in and booking ease, upgrade economy class comfort

and amenities, and reduce arrival delays. These represent the most controllable variables for improving overall passenger satisfaction.

6.6 Comparison with Prior Studies:

Test accuracy for logistic regression (84%) was below Xue (2025) at 85%, Bilgin (2025) at 87%, and Jiang et al. (2022) at 87%. Accuracy for gradient boosted trees (90%) fell below Bilgin's GBT (94%) and Jiang et al.'s (2022) random forest (96.3%). These differences are likely attributable in part to dataset size. Smaller training sets limit a model's ability to learn from the data, reducing predictive performance (Uddin & Lu, 2024). Xue (2025) used over 25,000 observations and Jiang et al. (2022) and Bilgin (2025) used over 100,000, compared to 5,196 in this study. Jiang et al. (2022) also applied RF-RFE recursive feature selection, reducing their feature set from 23 to 17 variables, with feature selection shown to improve accuracy by removing redundant and irrelevant features (Brzezinski, 2020). Noting that Jiang et al. (2022) was the only one of the three studies to report AUC-ROC, this study's LR AUC-ROC of 0.91 exceeded Jiang et al.'s value of 0.87, and GBT achieved a comparable AUC-ROC of 0.97 against Jiang et al.'s 0.96. Additionally, this study's GBT F1-score of 0.89 exceeded Xue's (2025) corresponding value of 0.87, while this study's LR F1-score of 0.82 matched Xue's (2025) value of 0.82. Both models remained marginally below Bilgin's (2025) GBT and LR F1-scores of 0.93 and 0.85, respectively. GBT is therefore the superior model, outperforming LR across all metrics while requiring no imputation, standardisation, or variable removal.

7. Conclusion and Reflections:

This study aimed to predict airline passenger satisfaction using binary classification and identify the key features influencing satisfaction, using a dataset of 5,196 observations. Logistic regression and gradient boosted trees were implemented in KNIME 5.3.3, with 5-fold cross-validation and hyperparameter tuning applied to both models. GBT outperformed LR across all metrics, achieving an AUC-ROC of 0.97 compared to LR’s 0.91. LR coefficient analysis identified travel type, cardholder status, and digital service quality as the strongest predictors of satisfaction.

Several methodological limitations warrant reflection. LR required removal of departure delay due to multicollinearity, imputation across several variables with substantial

missingness, and z-score standardisation, none of which were required by GBT. Mode and median imputation may have introduced bias into LR's coefficient estimates (Meng, 2025); multiple imputation would better preserve underlying data relationships (Meng, 2025). Additionally, associations between categorical predictors were not formally tested, meaning undetected multicollinearity may have further distorted LR's coefficients (Luqman et al., 2025). Finally, feature importance for GBT could not be displayed in KNIME 5.3.3, limiting deeper interpretation of that model's predictions.

The dataset also presents notable limitations. It excludes in-flight Wi-Fi service, ranked among the most influential predictors of satisfaction in comparable studies (Jiang et al., 2022; Xue, 2025), which may have limited model performance.

The LR coefficient analysis reveals that business travelers have 937% higher odds of satisfaction than personal travellers, and business class passengers are 108.1% more likely to be satisfied than economy passengers. This implies that passengers who are travelling for work or are wealthier receive better service. These are discriminatory patterns that algorithms trained on this data will learn to replicate and reinforce. Furthermore, demographic variables including gender and age were present in the dataset. While they were found to be not statistically significant predictors in this study, their inclusion raises fairness concerns. Predictive models that encode protected characteristics risk producing discriminatory outcomes (Mehrabi et al., 2022). Therefore, airlines deploying satisfaction prediction models should conduct fairness audits to ensure predictions do not systematically disadvantage passengers based on protected characteristics.

Future research could explore gradient boosting variants such as XGBoost or LightGBM on larger datasets, incorporate multiple imputation for LR, and apply Cramér's V to evaluate categorical multicollinearity prior to LR modelling.

References

Abedin, T., Xu, H., & Uddin, S. (2026). The impact of K selection in K-fold cross-validation on bias and variance in supervised learning models. *Scientific Reports*, *16*(1), Article 6084. https://doi.org/10.1038/s41598-026-37247-x

Alam, S., Ayub, M. S., Arora, S., & Khan, M. A. (2023). An investigation of the imputation techniques for missing values in ordinal data enhancing clustering and classification analysis validity. *Decision Analytics Journal*, *9*, Article 100341. https://doi.org/10.1016/j.dajour.2023.100341

Arafat, J., Tasmin, F., & Poudel, S. (2025). *Feature Selection and Regularization in Multi-Class Classification: An Empirical Study of One-vs-Rest Logistic Regression with Gradient Descent Optimization and L1 Sparsity Constraints*. https://doi.org/10.48550/arxiv.2510.14449

Arnold, C., Biedebach, L., Küpfer, A., & Neunhoeffer, M. (2024). The role of hyperparameters in machine learning models and how to tune them. *Political Science Research and Methods*, *12*(4), 841–848. https://doi.org/10.1017/psrm.2023.61

Arshad, M. A., Shahriar, S., & Anjum, K. (2023). *The Power Of Simplicity: Why Simple Linear Models Outperform Complex Machine Learning Techniques -- Case Of Breast Cancer Diagnosis*. ArXiv.org. https://doi.org/10.48550/arXiv.2306.02449

Bilgin, S. (2025). Prediction of Airline Passenger Satisfaction Using Machine Learning and Deep Learning Approaches. *2025 Innovations in Intelligent Systems and Applications Conference (ASYU)*, 1–7. https://doi.org/10.1109/asyu67174.2025.11208254

Brzezinski, D. (2020). *Fibonacci and k-Subsecting Recursive Feature Elimination*. https://doi.org/10.48550/arxiv.2007.14920

Chen, T., & Guestrin, C. (2016). XGBoost: a Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16*, 785–794. https://doi.org/10.1145/2939672.2939785

Das, U., Maiti, T., & Pradhan, V. (2010). Bias correction in logistic regression with missing categorical covariates. *Journal of Statistical Planning and Inference*, *140*(9), 2478–2485. https://doi.org/10.1016/j.jspi.2010.02.018

Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J. R. G., Gruber, B., Lafourcade, B., Leitão, P. J., Münkemüller, T., McClean, C., Osborne, P. E., Reineking, B., Schröder, B., Skidmore, A. K., Zurell, D., & Lautenbach, S. (2012). Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, *36*(1), 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x

Eshaghi, M. S., Afshardoost, M., Lohmann, G., & Moyle, B. D. (2024). Drivers and outcomes of airline passenger satisfaction: A Meta-analysis. *Journal of the Air Transport Research Society*, *3*, Article 100034. https://doi.org/10.1016/j.jatrs.2024.100034

Ge, Y., Li, Z., & Zhang, J. (2023). A simulation study on missing data imputation for dichotomous variables using statistical and machine learning methods. *Scientific Reports*, *13*(1), Article 9432. https://doi.org/10.1038/s41598-023-36509-2

Geron, A. (2023). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media, Inc. https://www.vlebooks.com/product/openreader?id=USheffield&accId=8860156&isbn=9781098122478

Greenwood, C. J., Youssef, G. J., Letcher, P., Macdonald, J. A., Hagg, L. J., Sanson, A., Mcintosh, J., Hutchinson, D. M., Toumbourou, J. W., Fuller-Tyszkiewicz, M., & Olsson, C. A. (2020). A comparison of penalised regression methods for informing

the selection of predictive markers. *PLOS ONE*, *15*(11), e0242730. https://doi.org/10.1371/journal.pone.0242730

Gudmundsson, S. V., Cattaneo, M., & Redondi, R. (2021). Forecasting temporal world recovery in air transport markets in the presence of large economic shocks: The case of COVID-19. *Journal of Air Transport Management*, *91*, Article 102007. https://doi.org/10.1016/j.jairtraman.2020.102007

Harrison, X. A., Donaldson, L., Correa-Cano, M. E., Evans, J., Fisher, D. N., Goodwin, C. E. D., Robinson, B. S., Hodgson, D. J., & Inger, R. (2018). A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ*, *6*(5), Article e4794. https://doi.org/10.7717/peerj.4794

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning, second edition : data mining, inference, and prediction*. Springer.

Hauke, J., & Kossowski, T. (2011). Comparison of Values of Pearson's and Spearman's Correlation Coefficients on the Same Sets of Data. *Quaestiones Geographicae*, *30*(2), 87–93. https://doi.org/10.2478/v10117-011-0021-1

Jäger, S., Allhorn, A., & Bießmann, F. (2021). A Benchmark for Data Imputation Methods. *Frontiers in Big Data*, *4*, Article 693674. https://doi.org/10.3389/fdata.2021.693674

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2014). *An introduction to statistical learning: with applications in R*. Springer. https://ebookcentral.proquest.com/lib/sheffield/detail.action?docID=6312402

Jiang, X., Zhang, Y., Li, Y., & Zhang, B. (2022). Forecast and analysis of aircraft passenger satisfaction based on RF-RFE-LR model. *Scientific Reports*, *12*(1), Article 11174. https://doi.org/10.1038/s41598-022-14566-3

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, *4*(9), Article 100804. https://doi.org/10.1016/j.patter.2023.100804

Laska, R. R., & Yolanda, A. M. (2024). A Comparative Study of Z-Score and Min-Max Normalization for Rainfall Classification in Pekanbaru. *Journal of Data Science*. https://doi.org/10.61453/jods.v2024no04

Li, J. (2024). Area under the ROC Curve has the most consistent evaluation for binary classification. *PLOS ONE*, *19*(12), Article e0316019. https://doi.org/10.1371/journal.pone.0316019

Li, K., Yao, S., Zhang, Z., Cao, B., Wilson, C. M., Kalos, D., Kuan, P. F., Zhu, R., & Wang, X. (2022). Efficient gradient boosting for prognostic biomarker discovery. *Bioinformatics*, *38*(6), 1631–1638. https://doi.org/10.1093/bioinformatics/btab869

Little, R. J. A., & Rubin, D. B. (2003). *Statistical analysis with missing data (second edition)*. John Wiley & Sons.

Luqman, M., Bhatti, S. H., Aydin, D., & Jamil, M. (2025). Addressing multicollinearity in general linear model: A novel approach for ridge parameter with performance comparison. *PLOS One*, *20*(10), Article e0335072. https://doi.org/10.1371/journal.pone.0335072

Maniwa, R., Ichijo, N., Nakahara, Y., & Matsushima, T. (2024). Boosting-Based Sequential Meta-Tree Ensemble Construction for Improved Decision Trees. *ArXiv* . https://doi.org/10.48550/arxiv.2402.06386

McHugh, M. L. (2012). Interrater reliability: the kappa statistic. *Biochemia Medica*, *22*(3), 276–282. https://doi.org/10.11613/bm.2012.031

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2022). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*, *54*(6), Article 115. https://doi.org/10.1145/3457607

Meng, H. (2025). A Comparative Study on Missing Value Imputation Techniques in Machine Learning. *SHS Web of Conferences*, *218*, 2014. https://doi.org/10.1051/shsconf/202521802014

Mittal, I., Kedia, A., Nalini, A., Sharma, A., Jain, S., & Gourisaria, M. K. (2026). Predicting Airline Passenger Satisfaction Using Machine Learning: A Comparative Evaluation. *2026 5th International Conference on Sentiment Analysis and Deep Learning (ICSADL)*, 1109–1115. https://doi.org/10.1109/icsadl67539.2026.11451771

Natekin, A., & Knoll, A. (2013). Gradient boosting machines, a tutorial. *Frontiers in Neurorobotics*, *7*, Article 21. https://doi.org/10.3389/fnbot.2013.00021

Oyedele, O. (2023). Determining the optimal number of folds to use in a K-fold cross-validation: A neural network classification experiment. *Research in Mathematics*, *10*(1). https://doi.org/10.1080/27684830.2023.2201015

Park, K., Cha, M., & Rhim, E. (2018). Positivity Bias in Customer Satisfaction Ratings. *Companion Proceedings of the the Web Conference 2018*, 631–638. https://doi.org/10.1145/3184558.3186579

Peng, J., Hahn, J., & Huang, K.-W. (2023). Handling Missing Values in Information Systems Research: A Review of Methods and Assumptions. *Information Systems Research*, *34*(1), 5–26. https://doi.org/10.1287/isre.2022.1104

Pinheiro, J. M. H., Oliveira, S. V. B., Silva, T. H. S., Saraiva, P. A. R., Souza, E. F., Godoy, R. V., Ambrosio, L. A., & Becker, M. (2025). The Impact of Feature Scaling In Machine Learning: Effects on Regression and Classification Tasks. *IEEE Access*, *13*, 199903–199931. https://doi.org/10.1109/access.2025.3635541

Ramadhan, N. G., & Putrada, A. G. (2023). XGBoost for Predicting Airline Customer Satisfaction Based on Computational Efficient Questionnaire. *International Journal on Information and Communication Technology (IJoICT)*. https://doi.org/10.21108/ijoict.v9i2.864

Ren, J., Ning, Z., Kirkness, C. S., Asche, C. V., & Wang, H. (2014). Risk of using logistic regression to illustrate exposure-response relationship of infectious diseases. *BMC Infectious Diseases*, *14*(1), Article 540. https://doi.org/10.1186/1471-2334-14-540

Rizkallah, L. W. (2025). Enhancing the performance of gradient boosting trees on regression problems. *Journal of Big Data*, *12*(1), Article 35. https://doi.org/10.1186/s40537-025-01071-3

Sakho, A., Malherbe, E., & Scornet, E. (2026, February 6). *Do we need rebalancing strategies? A theoretical and empirical study around SMOTE and its variantscomparisons of rebalancing strategies*. ArXiv.org. https://doi.org/10.48550/arXiv.2402.03819

Singh, V., Pencina, M., Einstein, A. J., Liang, J. X., Berman, D. S., & Slomka, P. (2021). Impact of train/test sample regimen on performance estimate stability of machine learning in cardiovascular imaging. *Scientific Reports*, *11*(1), Article 14490. https://doi.org/10.1038/s41598-021-93651-5

Soklaridis, S., Geske, A. M., & Kummer, S. (2024). Key characteristics of perceived customer centricity in the passenger airline industry: A systematic literature review. *Journal of the Air Transport Research Society*, *3*, Article 100031. https://doi.org/10.1016/j.jatrs.2024.100031

Sun, X., Wandelt, S., & Zhang, A. (2022). COVID-19 pandemic and air transportation: Summary of Recent Research, Policy Consideration and Future Research Directions. *Transportation Research Interdisciplinary Perspectives*, *16*, Article 100718. https://doi.org/10.1016/j.trip.2022.100718

Tharwat, A. (2021). Classification assessment methods. *Applied Computing and Informatics*, *17*(1), 168–192. https://doi.org/10.1016/j.aci.2018.08.003

Uddin, S., & Lu, H. (2024). Dataset meta-level and statistical features affect machine learning performance. *Scientific Reports*, *14*(1), Article 1670. https://doi.org/10.1038/s41598-024-51825-x

Xue, J. (2025). Airline Passenger Satisfaction Prediction and Key Influential Factors Identification Based on Logistic Regression and Random Forest. *Proceedings of the 2nd International Conference on Innovations in Applied Mathematics, Physics, and Astronomy*, 610–614. https://doi.org/10.5220/0013834200004708

Yildiz, A. Y., & Kalayci, A. (2025). *Gradient Boosting Decision Trees on Medical Diagnosis over Tabular Data*. https://doi.org/10.48550/arxiv.2410.03705

Zhang, J., & Yin, K. (2023). Application of gradient boosting model to forecast corporate green innovation performance. *Frontiers in Environmental Science*, *11*, Article 1252271. https://doi.org/10.3389/fenvs.2023.1252271

Zhang, L., Geisler, T., Ray, H., & Xie, Y. (2022). Improving logistic regression on the imbalanced data by a novel penalized log-likelihood function. *Journal of Applied Statistics*, *49*(13), 3257–3277. https://doi.org/10.1080/02664763.2021.1939662

Zhang, Y. (2024). Satisfaction analysis of airline passenger experience. *Proceedings of the 1st International Conference on Engineering Management, Information Technology and Intelligence*, 141–152. https://doi.org/10.5220/0012911400004508